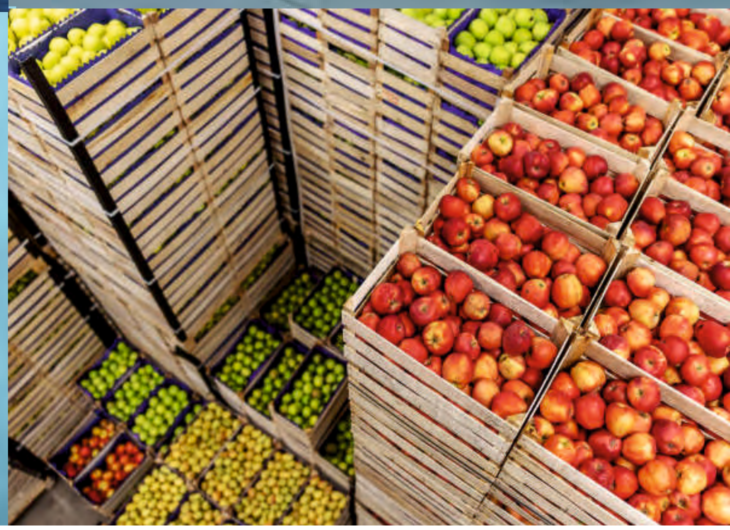
सत्यमेव जयते
Government of India

GUIDEBOOK ON

SUSTAINABLE TECHNOLOGIES FOR INFRASTRUCTURE RELATING TO COLD CHAIN SECTOR





OZONE CELL
Ministry of Environment, Forest and Climate Change
(MoEF&CC)

GUIDEBOOK ON

SUSTAINABLE TECHNOLOGIES FOR INFRASTRUCTURE RELATING TO THE COLD CHAIN SECTOR

SEPTEMBER 2024

सत्यमेव जयते

Government of India

OZONE CELL

Ministry of Environment, Forest and Climate Change (MoEF&CC)

Published by MoEF&CC

Acknowledgments

September 2024

MoEF&CC and PwC India are grateful to all experts and specialists for providing insights during the preparation of this report.

The team extends its sincere gratitude to:
- **Ms. Rajasree Ray,** Economic Advisor, Ministry of Environment, Forest and Climate Change
- **Mr. Aditya Narayan Singh,** Director (O), Ozone Cell (MoEF&CC).

The team also acknowledges the support provided by the various organizations and experts during the stakeholder consultation. We are thankful to the Ministry of Environment, Forest, and Climate Change (MoEF&CC), National Centre for Cold-Chain Development (NCCD), Department of Agriculture, Cooperation & Farmers Welfare, Ministry of Agriculture and Farmers Welfare, Ministry of Food Processing Industry, Bureau of Energy Efficiency (BEE), industry associations, academia, technology providers, consultants, and other experts for providing their inputs during the development of this guidebook.

This guidebook is developed as part of enabling activities of HPMP Stage-II project. Ozone Cell, Ministry of Environment, Forest and Climate Change (MoEF&CC) and United Nations Environment Programme (UNEP) are jointly implementing the enabling activities of HPMP Stage-II.

मंत्री
पर्यावरण, वन एवं जलवायु परिवर्तन
भारत सरकार

सत्यमेव जयते

भूपेन्द्र यादव
BHUPENDER YADAV

MINISTER
ENVIRONMENT, FOREST AND CLIMATE CHANGE
GOVERNMENT OF INDIA

MESSAGE

The cold chain logistics sector in India is witnessing significant growth, driven by the increasing demand for temperature-sensitive products, advancements in technology, and various initiatives including at the Government level.

The cold chain industry, by focusing on preserving product integrity, reducing food waste, and ensuring regulatory compliance, can significantly contribute to addressing sustainable development goals along with minimizing environmental impact and improving social and economic benefits. Towards this, there is a need to promote adoption of low global warming potential and energy efficient technologies, capacity building in good servicing practices of the cold chain equipment including effective handling of alternative low global warming potential refrigerants, besides logistics management.

The action points for the Cold Chain thematic area in the India Cooling Action Plan (ICAP) focusses on promoting adoption of energy efficient, non-ozone depleting substances and low global warming potential technologies in different components of cold chain, which will be a major driver in supporting the Government's priority areas of reducing food loss, ensuring food security and doubling farmers income.

The study report on sustainable technologies for new and retrofitting of existing cooling equipment / storage infrastructure for cold chain sector provides insights on critical areas including cooling technology, refrigerant choice, insulation, integration of renewable energy, and the operation and maintenance of the facility, which will act as crucial resource for stakeholders for enhancing awareness.

I congratulate all those involved in the preparation of this report.

(Bhupender Yadav)

पर्यावरण भवन, जोर बाग रोड़, नई दिल्ली-110003, फोन : 011-20819190, 011-20819187, फैक्स: 011-20819299
Paryavaran Bhawan, Jor Bagh Road, New Delhi-110003, Tel.: 011-20819190, 011-20819187, Fax : 011-20819299
ईमेल / E-mail : mefcc@gov.in

कीर्तिवर्धन सिंह
KIRTI VARDHAN SINGH

राज्य मंत्री
पर्यावरण, वन एवं जलवायु परिवर्तन
विदेश मंत्रालय
भारत सरकार
MINISTER OF STATE
ENVIRONMENT, FOREST AND CLIMATE CHANGE
EXTERNAL AFFAIRS
GOVERNMENT OF INDIA

Message

The rising temperature, increase in the incidence of hot and humid climatic conditions, country's growing population, economic growth have not only led to increase in demand for fresh and temperature sensitive products, but also challenges to the storage and transportation of these products. Cold chain storage helps to preserve the quality and shelf life of temperature-sensitive products by maintaining a desired low temperature range throughout the supply chain.

Refrigeration systems are important in the cold chain industry for storing perishable items and other temperature-sensitive products at a controlled low temperature. In these systems, choice of refrigerant depends on factors such as the size of the cold storage facility, the desired temperature range, safety considerations, energy efficiency, and compliance with regulations and standards. Developing sustainable cold storage infrastructure necessitates not only developing and promoting good management practices for cold storages, but also improve the operational efficiency of the cold storage infrastructure by adopting climate friendly refrigerants for the cooling systems.

Pursuant to the Kigali Amendment to the Montreal Protocol, cold storage facilities are transitioning towards more sustainable refrigerants, with an emphasis on alternatives with low/lower global warming potential to minimize adverse environmental impact.

The publication of the Guidebook on sustainable technologies for cold chain infrastructure is very timely and would serve as an important resource material and should be disseminated widely amongst all concerned stakeholders.

I complement the team associated in bringing out this publication.

(Kirti Vardhan Singh)

कार्यालय: 5वां तल, आकाश विंग, इंदिरा पर्यावरण भवन, जोर बाग रोड, नई दिल्ली-110003, दूरभाष: 011-20819418, 011-20819421, फैक्स: 011-20819207, ई-मेल : mos.kvs@gov.in
Office : 5th Floor, Aakash Wing, Indira Paryavaran Bhawan, Jor Bagh Road, New Delhi-110003, Tel.: 011-20819418, 011-20819421, Fax : 011-20819207, E-mail : mos.kvs@gov.in
कार्यालय: कमरा नं.141, साउथ ब्लॉक, नई दिल्ली-110001, दूरभाष: 011-23011141, 23014070, 23794337, फैक्स : 011-23011425, ई-मेल : mos.kvs@gov.in
Office : Room No. 141, South Block, New Delhi-110001, Tel. : 011-23011141, 23014070, 23794337, Fax : 011-23011425, E-mail : mos.kvs@gov.in
निवास: 23, बी.आर. मेहता लेन, नई दिल्ली-110001, दूरभाष: 011-23782979
Residence : 23, B.R. Mehta Lane, New Delhi-110001, Tel.: 011-23782979

लीना नन्दन
LEENA NANDAN

सत्यमेव जयते

सचिव
भारत सरकार
पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय
SECRETARY
GOVERNMENT OF INDIA
MINISTRY OF ENVIRONMENT, FOREST
& CLIMATE CHANGE

MESSAGE

The cold chain infrastructure is a vital part for the growth of agriculture and allied sectors. Agri produce, particularly those perishable in nature, require a temperature-controlled supply chain involving efficient storage, transportation, and distribution to increase the shelf life of the products. Apart from the agriculture and allied industries, cold chain infrastructure is also critical for food processing, vaccines and the chemical industry.

Improving the cold chains and post-harvest infrastructure will significantly reduce waste, create a surplus for exports, and enhance the income of farmers, which would significantly contribute to achieving the Sustainable Development Goal of zero hunger. The India Cooling Action Plan (ICAP), with cold chain as one of the thematic areas, has proposed specific action points towards promoting a sustainable cold chain infrastructure.

Considering the vast opportunities in the cold storage infrastructure, a multi-pronged approach is required to create modern infrastructure keeping in mind the requirements of various sectors such as climate friendly cooling systems, energy efficient design of cold storage infrastructure and enhancing capacities of personnel associated with operation and maintenance through development and implementation of a unified training and certification system for the service technicians.

The Guidebook on sustainable technologies for cold chain infrastructure addresses key issues for developing and management of sustainable cold chain infrastructure and will be a useful reference document for all the concerned stakeholders in this sector.

I take this opportunity to compliment the team involved in the preparation of this Guidebook.

September 6, 2024

(LEENA NANDAN)

इंदिरा पर्यावरण भवन, जोर बाग रोड़, नई दिल्ली-110 003 फोन : (011)-2081-9408, 2081-9308, फैक्स : (011)-2081-9238
INDIRA PARYAVARAN BHAWAN, JOR BAGH ROAD, NEW DELHI-110 003, PH. : 011-2081-9408, 2081-9308, FAX : 011-2081-9238
E-mail : secy-moef@nic.in, Website : moef.gov.in

LiFE Lifestyle for Environment

Contents

Figures and Tables

List of Figures

List of Tables

Abbreviations

AC	Alternating Current
AMF	Automatic Mains Failure
ARS	Adsorption Refrigeration System
ASHRAE	American Society of Heating, Refrigerating, and Air-Conditioning Engineers
BCG	Bacillus Calmette-Guérin
BEE	Bureau of Energy Efficiency
BIS	Bureau of Indian Standards
CAPEX	Capital Expenditure
CFB	Corrugated Fiber Board
CFC	Chlorofluorocarbon
CoP	Coefficient of Performance
DC	Direct Current
DF	Deep Freezer
DG	Diesel Generator
DTP	Diphtheria-Tetanus-Pertussis
DVS	District Vaccine Stores
DX	Direct Expansion
EEV	Electronic Expansion Valve
EPS	Expanded Polystyrene
GI	Galvanized Iron
GMSD	Government Medical Store Depot
GWP	Global Warming Potential
HCFC	Hydrochlorofluorocarbon
HFC	Hydrofluorocarbon
HFO	Hydrofluoroolefin
HPMP	HCFC Phaseout Management Plan
ICAP	India Cooling Action Plan
IE	International Efficiency
ILR	Ice Lined Refrigerator
INR	Indian Rupee
ISO	International Organization for Standardization

ISO/TC	International Organization for Standardization Technical Committee
JE	Japanese Encephalitis
LN2	Liquid Nitrogen
MIDH	Mission for Integrated Development of Horticulture
NHB	National Horticulture Board
O&M	Operations & Maintenance
ODP	Ozone Depleting Potential
OPEX	Operational Expenditure
OPV	Oral Polio Vaccine
PCM	Phase Change Material
PFAS	Perfluoroalkyl and Polyfluoroalkyl Substances
PHC / CHC	Primary Health Centre / Community Health Centre
PHE	Plate Heat Exchanger
PIR	Polyisocyanurate
PLC	Programmable Logic Controller
PoE	Polyolester
PPA	Power Purchase Agreements
PUF	Polyurethane Foam
RE	Renewable Energy
REC	Renewable Energy Certificates
RVS	Regional Vaccine Store
S&L	Standards & Labelling
SVS	State Vaccine Store
TR	Tonnes of Refrigeration
TT	Tetanus Toxoid
TXV	Thermostatic Expansion Valve
ULT	Ultra Low Temperature
USD	United States Dollar
VAM	Vapor Absorption Machine
VCM	Vapor Compression Machine
VFD	Variable Frequency Drive
WIC	Walk-In-Chiller
WIF	Walk-In-Freezer
XPS	Extruded Polystyrene

1

Navigating this Guidebook

1.1. Purpose

The objective of this guidebook is to assist stakeholders within the cold chain sector to make informed decisions regarding sustainable interventions in their cold chain facilities, whether they are considering retrofitting existing structures or establishing new ones. This guidebook provides thorough insights into critical areas including cooling technology, refrigerant choice, insulation, integration of renewable energy, and the operation and maintenance of the facility. This is to enable stakeholders to move towards more sustainable practices. By offering pragmatic guidance, the guidebook endeavours to promote energy-efficient cold chain solutions, thus ensuring the integrity of perishable goods and vaccines, whilst concurrently reducing the industry's carbon footprint.

1.2. Targeted Audience/Stakeholders

This guidebook has been meticulously crafted to cater to a diverse array of stakeholders, all of whom hold pivotal positions within the cold chain sector. The content has been specifically tailored to meet their distinct needs and interests, as delineated below.

Facility Managers/Technicians:

Facility managers are the hands-on leaders tasked with the day-to-day operations of cold chain facilities. They are responsible for overseeing the management of refrigeration systems, storage, and transportation.

How This Guidebook Benefits Them: Facility managers will discover practical insights and best practices to implement within their operations.

Business Owners and Entrepreneurs:

Owners of cold chain facilities, large and small, as well as entrepreneurs aspiring to enter the market, are crucial stakeholders.

How This Guidebook Benefits Them: This guidebook offers invaluable information for making informed decisions about investments in energy-efficient cold chain infrastructure. It delineates the current technologies in use and potential sustainable technologies that can be integrated into their cold chain infrastructure.

Policymakers and Government Agencies:

Policymakers and government agencies are responsible for overseeing regulations, standards, and incentives within the cold chain sector. They have a pivotal role in steering the industry's sustainability efforts.

How This Guidebook Benefits Them: The guidebook can assist policymakers in making informed decisions regarding the development of regulations and incentives that encourage sustainable cold chain practices. It offers a comprehensive overview of the technologies available.

Investors and Financial Institutions:

Investors and financial institutions provide the necessary funding and resources for cold chain projects. They have an increasing interest in environmentally responsible investments.

How This Guidebook Benefits Them: Investors can utilise this guidebook to evaluate the sustainability of cold chain projects and to understand the potential risks and rewards associated with various technologies and practices.

1.3. Scope

- Comprehensive Coverage: The guidebook provides comprehensive coverage of sustainable technologies and practices within the perishables and vaccines cold chain sector. It encompasses key areas such as cooling technology, refrigerant selection, insulation, RE (Renewable Energy) integration, and O&M (Operations & Maintenance) practices.

- Practical Guidance: The guidebook goes beyond theory and offers practical guidance. It includes good practices to illustrate how stakeholders can implement sustainable solutions effectively.

1.4. How to use this Guidebook

Navigate through this guidebook seamlessly as it delves into two critical sections of the cold chain: perishables cold chain and the vaccine cold chain. The first chapter in each section—perishables and vaccine cold chain—provides an overview of current cooling technologies, refrigerants, insulation, and O&M practices.

Subsequently, the remaining chapters are dedicated to addressing packhouses, refrigerated vehicles, cold storages, and ripening chambers for the perishable cold chain, and refrigerated vehicles and cold storages for the vaccine cold chain, respectively. Each chapter introduces their role in the cold value chain. Within each chapter, insights on retrofitting options, encompassing cooling technology, insulation, RE integration, and refrigerants are provided, followed by factors for consideration for a new setup of a cold chain facility.

For the realisation of your project, whether it involves a new setup or retrofitting, consider collaborating with experts or consultants based on the specific requirements of the relevant segment. Engage in a feasibility analysis with their assistance to ensure informed decision-making throughout the project.

2

Perishables Cold Chain

2.1. Overview of the existing landscape of perishable cold chain

In the perishables cold chain, maintaining product quality is of the utmost importance. This begins with swift transportation from farms to packhouses, where processes such as washing, drying, sorting, grading, and precooling take place. Precooling is essential for the removal of field heat, as it extends the shelf life and preserves the nutritional value of the produce. Subsequently, the materials are moved to cold storage units, which are specifically designed to meet the requirements of the supply chain. Here, temperatures are meticulously regulated to inhibit the growth of microorganisms and to decelerate the ripening process. For certain products, such as bananas, ripening chambers are utilised to ensure the development of optimal flavour and texture. Refrigerated transportation, whether by container or lorry, maintains the necessary temperatures, thus guaranteeing the freshness of the products. This carefully orchestrated cold chain reaches its conclusion at retail points, where it provides consumers with access to safe, high-quality perishables and concurrently minimises waste. Given the energy-intensive nature of cooling, the integration of sustainable practices is imperative. Such measures serve to reduce carbon emissions and enhance the overall energy efficiency of the cold chain system.

Figure 1 Perishables cold chain [1]

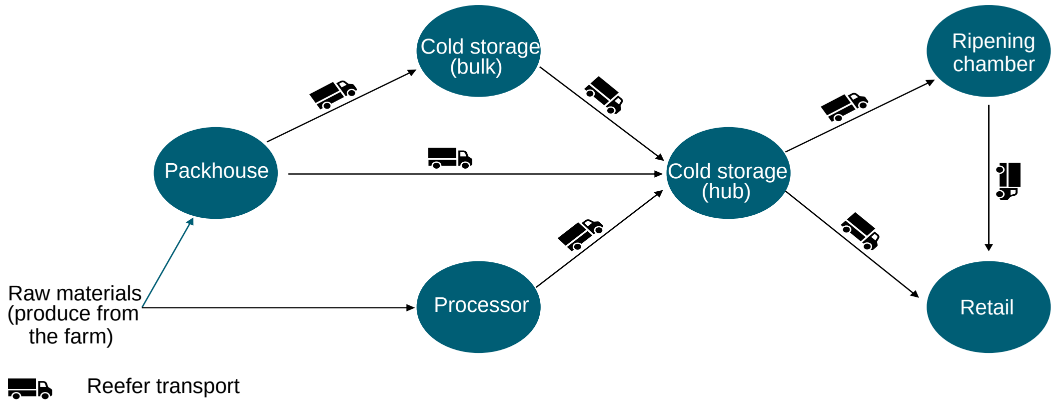


Image source: India Cooling Action Plan (ICAP)

Current infrastructure:

In the Indian cold chain sector, predominant attention has been directed towards the development of cold storages, wherein the infrastructure is relatively robust with a less than 10% gap in current capacities. However, a substantial disparity exists in the infrastructure of other vital cold chain segments. According to the ICAP (India Cooling Action Plan), packhouses exhibit a considerable 99% gap, highlighting a significant deficiency in the current infrastructure. With approximately 500 packhouses in India at present, the number is likely to increase to 55,000 in the next decade and to 125,000 in the subsequent decade, attributing to an energy consumption of 2.4 TWh and 5.2 TWh respectively.

Similarly, ripening chambers face a noteworthy 90% gap, and it is anticipated that there would be a steady growth in ripening chambers from current 1000 units to 9,000 units in the next decade and to 14,000 units in the subsequent decade.

Refrigerated vehicles, commonly referred to as ‘reefer vehicles’, currently exhibit a substantial shortfall, with a 85-90% gap between the extant infrastructure and the actual demand. The expansion of reefer vehicles is closely associated with the rise in the number of packhouses. It is projected that the number of units will increase to 135,000 in the forthcoming decade and to 400,000 in the decades thereafter, up from the current figure of 15,000 units.

Standards published by ISO/TC315 in the field of cold chain sector:

Below are a few standards that may be considered for adoption by industries for sustainable practices within the cold chain sector. Nevertheless, as new revisions of these standards emerge and additional standards are introduced, it is advisable to consult the ISO standards as a matter of routine practice within the industry.

Table 1 ISO standards for cold chain sector [2]

Published standards	Title
IS 10594: 2021	Specification for Thermostatic Expansion Valves
IS 10617: 2018	Hermetic compressors – Specification (Second Revision)
IS 11327: 2022	Requirements for Refrigerants Condensing Units (Second Revision)
IS 16590: 2017	Water cooled chilling packages using the vapour compression cycle – Specification
IS 16656: 2017 **ISO 817: 2014**	Refrigerants – Designation and safety classification
IS 16678 (Part 1 to 4): 2018 **ISO 5149 Pt 1 to 4:2014**	Refrigerating Systems and Heat Pumps – Safety and Environmental Requirements: Part 1 Definitions, Classification and Selection Criteria Part 2 design, construction, testing, marking and documentation. Part 3 installation site Part 4 operation, maintenance, repair and recovery
IS/ISO 17584: 2022 IS/ ISO 17584	Refrigerant properties (First Revision)
IS 17773: 2022	Closed-Circuit Ammonia Refrigeration System Code of Practice for Design and Installation
IS 18802: 2023 **ISO 22042:2021**	Blast chiller and freezer cabinets for professional use; Classification requirements and test conditions
IS 7872: 2020	Deep Freezers – Specification (Second Revision)

Growth scenario:

The perishables cold chain sector in India is poised for substantial growth, driven by an increasing demand for fresh produce, dairy, and processed food. As consumer expectations for quality and safety rise, there is a heightened focus on expanding and modernising cold storage facilities, packhouses, ripening chambers, and transportation infrastructure. Government initiatives supporting agricultural and cold chain development further contribute to a positive growth scenario. The integration of technology, the adoption of sustainable practices, and collaborations across the supply chain are expected to play pivotal roles in enhancing the efficiency and resilience of the perishables cold chain in India, creating new opportunities for stakeholders in the forthcoming years.

2.1.1. Cooling technologies

The two technologies predominantly in use are:

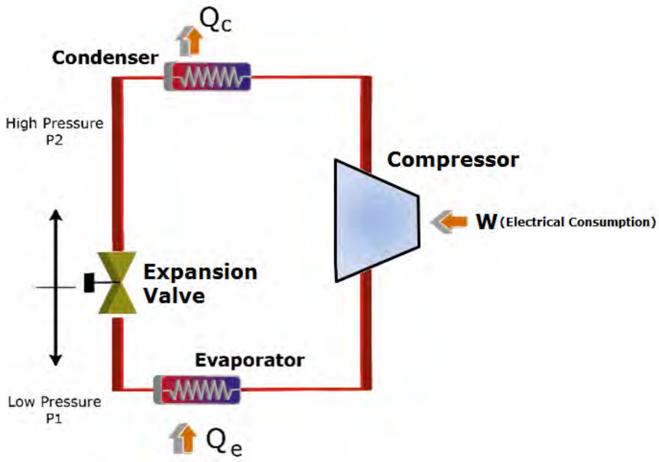
A. Vapor Compression Technology

B. Vapour Absorption Technology

A. Vapour Compression Technology:

The vapour compression system stands as the most widely utilised cooling technology in refrigeration applications. It operates on the principle of electricity-driven compressor machines. In this system, the refrigerant undergoes a cyclic process: commencing in the evaporator, where it absorbs heat and transitions to the compressor, undergoing adiabatic compression to high temperature and pressure. The compressed refrigerant then moves to the condenser for cooling at constant pressure. Subsequently, it is expanded through an expansion valve, leading to a decrease in both temperature and pressure. The evaporator, operating at constant pressure, absorbs heat, facilitating the cooling of the surrounding space. This cycle perpetuates itself, forming the core mechanism of the vapour compression system. In Figure 2, Qe and Qc stand for the heat absorbed and heat released by the refrigerant respectively.

Figure 2: Vapor compression cycle [3]



Note: *BEE (Bureau of Energy Efficiency) is presently working to develop voluntary S&L (Standards and Labelling) program for refrigerant compressors (having capacity between 1.5 kW – 15 kW) used in cold chain applications. It would help the stakeholders in making informed decisions during equipment selection across cold rooms, packhouses, ripening chambers, and reefer transport. BEE's S&L programme aims to help consumers make informed choices considering cost-effectiveness and energy performance of various energy-consuming appliances, thus enabling them to save energy, reduce electricity consumption and also contribute to a greener planet.*

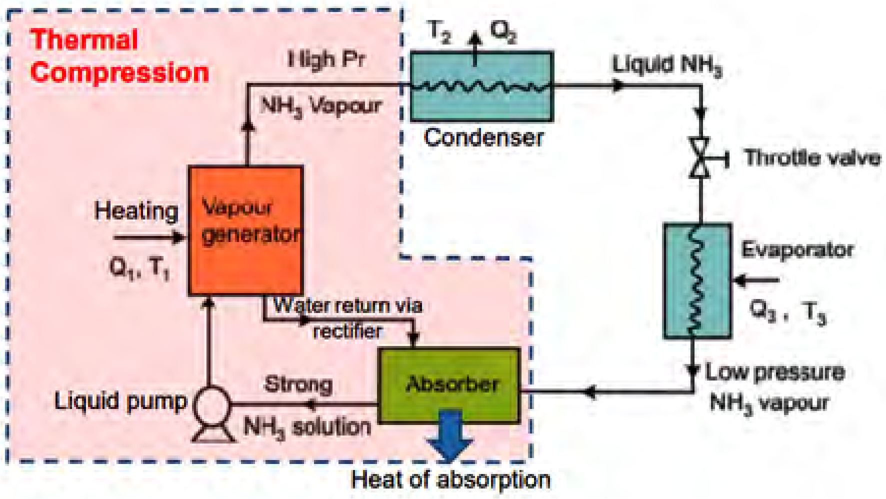
B. Vapour Absorption Refrigeration Technology:

Vapour absorption is an energy-efficient approach to cooling systems. In contrast to the traditional vapour compression system that relies on mechanical compressors, this system utilises a chemical absorption process to achieve cooling. The primary components comprise an absorber, generator, pump, and evaporator.

In vapour absorption technology, a liquid refrigerant, typically water, is absorbed by a suitable refrigerant, often ammonia or lithium bromide. The absorption of the refrigerant takes place in an absorber, resulting in a dilute solution. This solution is then pumped into a generator, where heat is applied to separate the refrigerant from the absorbent, producing a concentrated solution and a vaporised refrigerant. The vaporised refrigerant is subsequently condensed, and the cycle repeats. Unlike the vapour compression system, vapour absorption utilises heat as the primary energy input, rendering it suitable for applications where waste heat or alternative energy sources are available.

Q_1, Q_2 and Q_3 and T_1, T_2 and T_3 in Figure 3 represent the heat absorbed / released and the temperature during the process.

Figure 3: Vapor absorption cycle [4]



Despite its eco-friendly attributes, only 5-7%[5] of applications in India utilise vapour absorption technology, which underscores a substantial knowledge gap among stakeholders concerning its benefits, particularly when conditions favour its implementation. The ammonia-water-based absorption system, employing ammonia as the refrigerant and water as the transport medium, is the most prevalent in India. It is noteworthy that, in comparison to the lithium-bromide-based system, the ammonia-water system typically demonstrates lower efficiency with a reduced CoP (Coefficient of Performance) in refrigeration.

2.1.2. Refrigerants

Refrigerants are essential for the cooling and temperature regulation of a wide range of perishables and vaccines, which is vital for preserving the quality and safety of these products. Nevertheless, the environmental impact of refrigerants is an increasing concern. The ODP (Ozone Depleting Potential) and GWP (Global Warming Potential) serve as principal indicators of a refrigerant's environmental footprint.

It is imperative to transition to refrigerants with low or zero ODP and GWP, such as HFOs (Hydrofluoroolefins) and natural refrigerants, to alleviate environmental damage. The table below presents the refrigerants currently utilised alongside their replacement options, taking into account the environmental repercussions.

Table 2: Refrigerants and their GWP value [6]

Refrigerant	Type	GWP value (100 years)
CFC		
R-12	CFC	10910
R-502	CFC/HCFC Azeotrope	4657
HCFC		
R-22	HCFC	1810
HFC		
R-507	HFC	3985
R-404A	HFC	3920
R-428A	HFC	3607
R-434A	HFC	3245
R-422D	HFC	2729
R-424A	HFC	2440
R-407A	HFC	2110
R-410A	HFC	2088
R-442A	HFC	1888
R-407C	HFC	1770
R-453A	HFC	1765
R-426A	HFC	1508
R-134a	HFC	1430
R-32	HFC	675
HFO / HFO Blends		
R-464A	HFO Blend	1288
R-449A	HFO Blend	1282
R-448A	HFO Blend	1273
R-513A	HFO	573
R-450A	HFO Blend	547
R-454C	HFO	148
R-455A	HFO Blend	146

Refrigerant	Type	GWP value (100 years)
R-1336mzz	HFO	18
R-1234yf	HFO	4
R-1233zd	HFO	1
R-1234ze	HFO	0
Hydrocarbons and natural refrigerants		
R-170	Hydrocarbon	6
R-290 (Propane)	Hydrocarbon	3
R-600a	Hydrocarbon	3
R-744 (Carbon dioxide)	Natural	1
R-717 (Ammonia)	Natural	0

The relationship between the ODP and GWP of refrigerants and their efficiency is both complex and significant. Typically, refrigerants with low ODP and GWP are preferred due to their reduced environmental impact; however, this does not guarantee optimal efficiency. The CoP is the measure of efficiency for refrigerants.

Traditional Hydrochlorofluorocarbons (HCFCs) and Hydrofluorocarbons (HFCs), which have been utilised in the cold chain sector for many years, have demonstrated commendable efficiency. Consequently, they have been the predominantly used refrigerants within the Freon category. Freons with low ODP and GWP, such as HFOs, have not exhibited efficiency comparable to that of the traditional HCFCs and HFCs. This is the reason why blends have been developed that aim to reduce ODP and GWP while enhancing efficiency.

In the case of natural refrigerants, ammonia has negligible ODP and GWP values, as well as high performance efficiency. As a result, it is one of the most extensively used refrigerants in the cold chain sector in India. Additionally, the natural refrigerant R-290 (propane) has also begun to be employed in commercial last-mile applications by manufacturers, where the units are self-contained, and factory charged and sealed.

It is, therefore, imperative to find an equilibrium between environmental impact and efficiency when selecting the most appropriate refrigerant for a given application.

2.1.3. Thermal Insulation

Thermal insulation is an extremely important component in a cold chain system as it impacts the energy efficiency of a cold storage facility. It has two main functions to perform:

- To minimise the ingress of heat from the ambient to the inside cold space
- To minimise the ingress of moisture from the surrounding areas into the cold chambers

In recent times, there has been a significant shift within the insulation industry from traditional materials such as Expanded Polystyrene (EPS), Extruded Polystyrene (XPS), and mineral wool to the adoption of Polyurethane Foam (PUF) and Polyisocyanurate (PIR). This transition is propelled by the superior insulating properties provided by PUF and PIR, which mirror a wider industry trend towards enhancing thermal efficiency in construction materials. The table below compares the insulating properties of the materials currently utilised in the cold chain industry.

The quality of insulation can be determined through its U value, R value and K value. U value is defined as the thermal transmittance of the materials or the overall heat transfer coefficient which is the ability of the insulating material to transmit heat. The lower the U value, the better the insulation is. R value is defined as the thermal resistance of the insulation, which is the ability of the insulating material to resist heat flow. The higher the R value, the better the insulation is. Lastly, K value is defined as the thermal conductivity of the insulating material, which is the ability of the insulating material to conduct heat. The lower the K value, the better the insulation is.

Table 3: Insulation parameters [7,8,9,10,11,12,13,14]

Insulation material	U value	R value	K value
EPS	0.038 W/m²K	4 per inch of thickness	0.033 W/mK
XPS	0.033 W/m²K	4.7 per inch of thickness	0.024 W/mK
PUF	0.022 W/m²K	5.6-7 per inch of thickness	0.023 W/Mk
PIR	0.022 W/m²K	7-7.2 per inch of thickness	0.022 W/mK

2.1.3.1. Vapour barriers

Even if insulation is applied to a cold body, condensation will persist, as water vapour can permeate through the joints of the insulation. This compromises the efficacy of the insulation material and results in an elevated energy consumption for cooling. Consequently, a vapour barrier becomes an indispensable element of the cold chain. It is affixed atop the insulation joint on the warm side, preventing the ingress of water vapour or humidity. In conjunction with the vapour barrier, the correct application of adhesive and the securement of insulation within a supporting framework are paramount. These measures ensure that the insulation remains intact for an extended duration and functions efficiently.

Table 4: Vapour barrier classes [15]

Class	Materials
Class I (permeability: <=0.1)	Glass, sheet metal, polyethylene sheet, rubber membrane
Class II (Permeability: <0.1<=1.0)	Unfaced EPS/XPS, 30-pound asphalt coated paper, plywood, bitumen coated kraft paper
Class III (Permeability: <1.0<=10)	Gypsum board, fiberglass insulation (unfaced), cellulose insulation, board lumber, concrete block, 15-pound asphalt coated paper, house wrap

The higher the permeability value, the more vapor permeable the material becomes. Therefore, it is advisable to select a vapour barrier with a low permeability.

2.1.4. O&M Practices

In the cold chain sector, effective O&M practices play a crucial role in guaranteeing the consistent and efficient operation of refrigeration systems, storage facilities, and reefer transports.

However, in India, these practices encounter notable challenges that impede their strict implementation.

Factors such as inadequate awareness and education regarding optimal practices, insufficient training of personnel, and a fragmented cold chain infrastructure contribute to the lack of stringency in O&M standards when we talk about perishables (fruits and vegetables).

Need for streamlining of decommissioning of old systems:

Streamlined decommissioning of old systems in a cold chain is crucial for maintaining efficiency and safety. As technology evolves, outdated equipment can lead to inefficiencies, increased energy consumption, and potential risks to product quality and safety.

2.2. Packhouses

2.2.1. Introduction

A packhouse serves as the initial stage in the cold chain for perishables, where agricultural produce is transported directly from the farm. Within these facilities, processes such as sorting, grading, washing, drying, and weighing are conducted before the produce is dispatched to a cold storage facility. Traditionally, packhouses encompassed only the aforementioned activities.

However, the significance of incorporating a pre-cooling system within packhouses has escalated. This system is instrumental in extracting field heat from freshly harvested produce, thereby prolonging its shelf life and ensuring the preservation of its quality and nutritional value in preparation for the extended cold chain. Subsequently, the produce is placed in a staging cold room—also an integral component of the packhouse—where it is temporarily stored in a controlled environment prior to further transportation within the supply chain, whether to a local market, a bulk cold storage, or a processing facility. It is advantageous for these integrated packhouses, equipped with pre-cooling infrastructure and a staging cold room, to be situated in proximity to the farm gate to maximise their benefits.

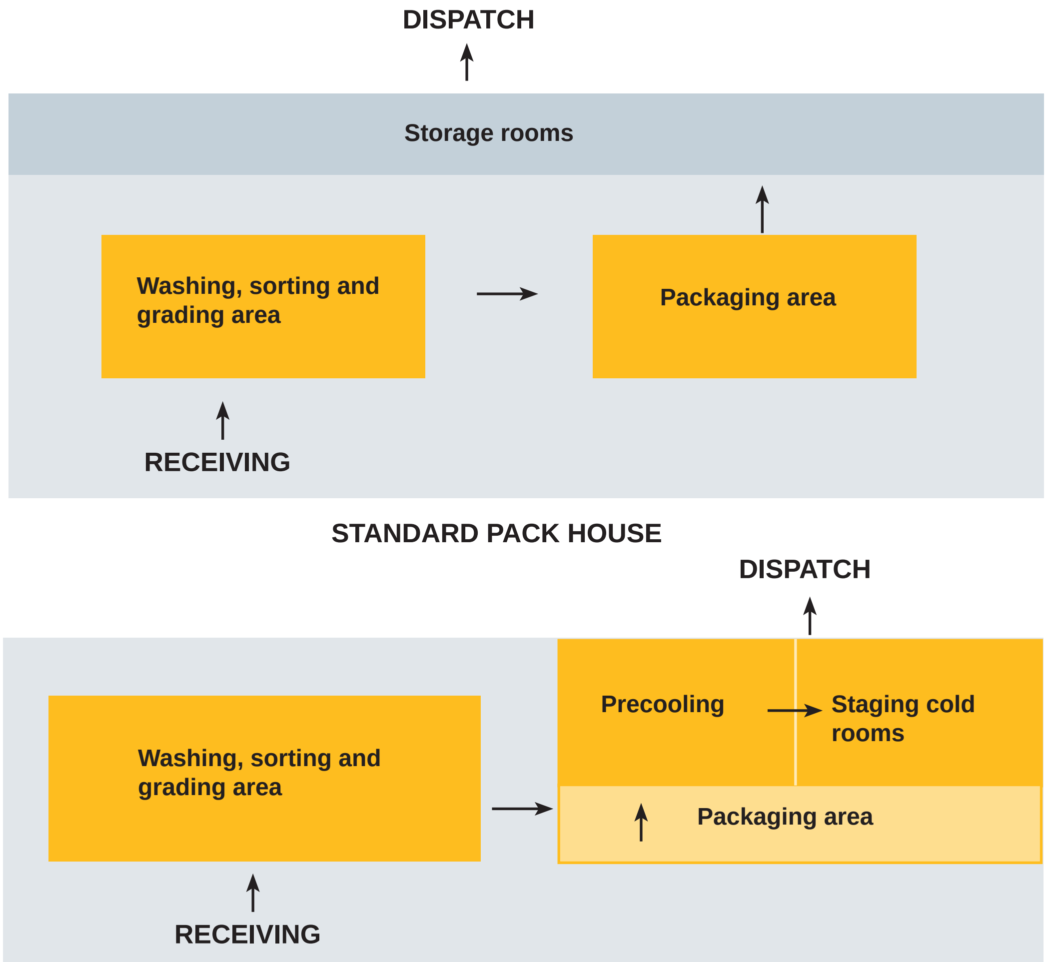
- Standard pack houses
- Integrated pack houses

Energy efficiency in packhouses can be achieved through a series of steps involving planning, designing, and operations. The design plays a critical role. As seen in figure 4, the planning, design, and layout can vary from packhouse to packhouse, depending on the type of produce which is generally stored. The following sections talk about ways of achieving energy efficiency.

2.2.2. Transitioning to sustainable options

Transitioning to sustainable options in packhouses within the cold chain involves a

Figure 4: Standard and integrated packhouse



comprehensive approach to reduce environmental impact and enhance efficiency, applicable to both, retrofitting existing facilities and designing new setups.

2.2.2.1. Retrofitting

Below are the potential retrofits in the following areas:

1. Insulation
2. Cooling technology
3. Refrigerant
4. RE integration
5. Other design interventions

1. Insulation

Polyurethane Foam (PUF) and Polyisocyanurate (PIR) emerge as recommended solutions due to their superior insulation properties in comparison to Expanded Polystyrene (EPS), Extruded Polystyrene (XPS), and mineral wool.

The requisite thickness of the insulation is determined on a case-by-case basis, taking into account the insulating material, the specific application, and the location of the facility. It is also imperative to ensure that the installation of insulation is executed meticulously, sealing all corners and joints comprehensively to prevent any ingress of ambient air into the room.

It is advised to consult a refrigeration expert or a consultant for economic feasibility prior to initiating any retrofit.

Retrofit option: EPS/XPS/Mineral wool to PUF/PIR

Retrofitting the insulation of existing infrastructure within cold chain represents a straightforward method for improving energy efficiency. Options such as transitioning from EPS or XPS or Mineral wool to Polyurethane PUF or PIR should be considered for their superior thermal performance. This enhancement reduces thermal ingress into the cold storage facility, consequently diminishing energy consumption and operational costs.

Please refer to **section 2.4.2.1 – Insulation,** for more details.

> **Note:** *A vapour barrier coating should be applied over PUF insulation. While PUF is exceptional in its thermal insulation properties, it is susceptible to moisture ingress, which can compromise its efficiency over time. The vapour barrier acts as a protective shield, preventing moisture infiltration and safeguarding the insulation from potential degrade.*

2. Cooling technology

Retrofit option 1: VFDs in a fixed speed compressor

The compressor can be fitted with an inverter-based VFD as well to improve the energy efficiency. More on this is discussed in the Cold storage chapter.

Please refer to **section 2.4.2.1 - Cooling technology** for more details.

Retrofit option 2: Higher efficiency compressor motor

Please refer to **section 2.4.2.1 - Cooling technology** for more details.

3. Refrigerant

Please refer to chapter **section 2.4.2.1** – Refrigerant for more details.

> **Note:** *Retrofitting from R-22 to R-134a ought not to be undertaken in the case of a precooling chamber. The rationale behind this is that a precooling chamber necessitates the reduction of temperature rather than merely maintaining it.*

It is advised to consult a refrigeration expert prior to initiating any retrofit.

4. RE integration

The integration of potential renewable energy sources with packhouses ensures the sustainable operation of refrigeration systems, where the dependency on the electricity grid is reduced, thereby diminishing the carbon footprint. Although numerous renewable energy sources are available, such as wind energy, geothermal energy, and solar energy, only solar has proven to be economically and practically feasible where applicable, with geothermal energy showing relatively lesser feasibility within the cold chain sector.

The most crucial factor to consider when contemplating the integration of a solar energy source into any facility is the availability of stable and continuous grid electricity. As most packhouses are situated at the farm gate level, there is often an unstable supply of electricity, rendering solar integration infeasible for providing cooling for the required duration.

In such instances, there is, however, the possibility of installing a diesel generator (DG) as a backup, with the DG being capable of handling the complete refrigeration load of the precooling unit of the packhouse. Nevertheless, the installation of such a substantial capacity DG incurs a very high capital expenditure cost, resulting in extended payback periods and rendering the system economically unviable.

In circumstances where a consistent and reliable grid supply is established, the potential for seamlessly incorporating solar solutions presents itself as a strategic avenue. For further details on solar integration in areas with adequate grid connectivity, please refer to **Section 2.4.2.1 – RE integration.**

It is advised to consult an RE integration expert for checking the economic feasibility prior to initiating any retrofit.

5. Other design interventions

Retrofit option 1 : Flush doors for the pre-cooling chambers:

Flush doors are a practical choice for precooling rooms and staging cold rooms in a packhouse facility because they can be designed with airtight seals and are highly durable. They have a tight closing mechanism which ensures that the doors are tightly sealed at all times.

Figure 5: Flush door [16]

Flush doors are recommended when the operation of the door is high.

Kick plates are generally used additionally as a protective metal attached to the bottom of the door.

Cost estimate

Table 5: Flush door costing [17]

Approximate temperature range (oC)	*Cost with one side kick plate (INR)	*Cost with two side kick plate (INR)
0 to +8	21,000	23,000
-5 to 0	26,000	27,500
-10 to -5	28,500	29,000
-25 to -10	31,000	31,500
-35 to -25	33,000	34,000

**This is an approximate rate. Cost of side kick plates will vary from manufacturer to manufacturer.*

Retrofit option 2: Dock shelters and dock levellers

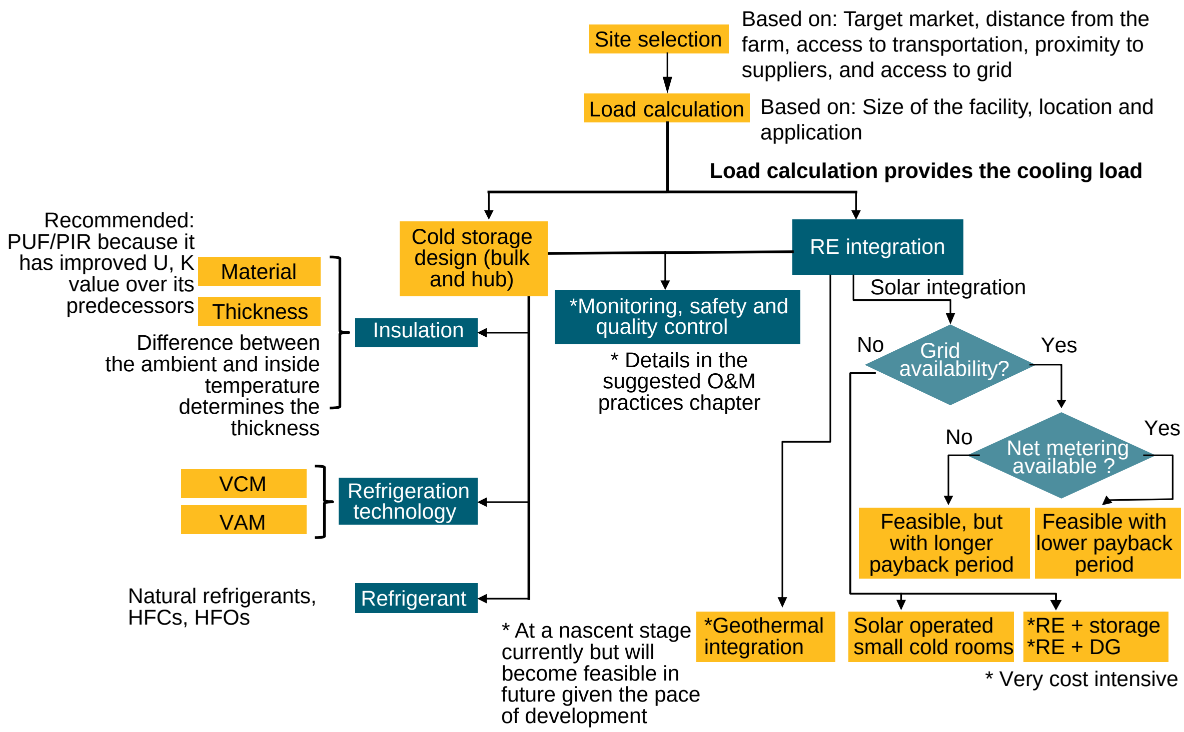
Retrofitting dock shelters and dock levellers in precooling and staging cold rooms within a packhouse can yield significant benefits. By creating a sealed environment around loading docks, dock shelters facilitate the maintenance of precise temperature control during the loading and unloading of perishable produce. This minimises the exchange of external air, enhancing energy efficiency and preventing temperature fluctuations. The retrofit ensures that precooling and staging cold rooms maintain the optimal conditions required for preserving the quality and freshness of fruits and vegetables. Furthermore, dock shelters act as barriers against contaminants, contributing to the hygiene standards in the packhouse. The improved temperature management and operational efficiency render dock shelters a practical solution for upgrading existing facilities, ensuring the seamless handling of produce while reducing energy consumption and preserving product integrity.

For details regarding the cost, please refer to **section 2.4.2.1 – Other design interventions**.

It is advised to consult a refrigeration expert prior to initiating any retrofit.

2.2.2.2. New setup

Figure 6: New setup for a packhouse



For the precooling chambers and staging cold rooms within a packhouse setup, the provided flowchart serves as a valuable reference for decision-making in the establishment of a new system. Site selection is the initial criteria, which depends on several factors like distance from the farm, access to transportation, target market, proximity to suppliers and access to grid. Following heat load calculations, the cooling technology, refrigerant selection, selection of the insulating material takes place.

Refer to **section 2.4.2.2.** for more details regarding usage of CO_2 as a refrigerant.

Also, RE integration feasibility can be assessed based on the location and feasibility analysis. Post the design of the precooling and staging cold rooms, monitoring, safety, and quality control measures should be defined, followed by documentation and compliance, overall ensuring an optimum functioning of the facility in terms of energy usage and sustainability.

It is advised to consult a refrigeration expert / consultant for setting up of a new packhouse for proper guidance and calculations.

2.2.3. Suggested O&M practices

Table 6: Suggested O&M practices for a packhouse

Segment	Suggested Practices
Equipment maintenance	Check the functioning of all components of refrigeration system/equipment regularly to prevent build-up of dirt, debris
Equipment calibration	Calibrate all the necessary measuring equipment regularly to ensure accurate readings and maintain the desired conditions for precooling and staging cold rooms.
Airflow optimisation	Ensure proper airflow through the pre-cooling system by inspecting and cleaning the air intake and exhaust components.
Routine inspection of doors and seals	Regularly inspect and maintain integrity of precooling and staging cold room doors and seals in order to prevent the ingress of warm air.
Temperature & Relative Humidity (RH) monitoring	Implement a robust monitoring system to track temperature humidity levels during the pre-cooling process.

Segment	Suggested Practices
Employee training	Train the respective personnel on proper pre-cooling procedures, including equipment operation, monitoring, and basic troubleshooting.
Documentation and compliance	Maintain appropriate temperature and RH monitoring logs along with the P&I diagrams and any other documentation required for compliance.
Emergency preparedness	Develop and regularly update emergency response plans. Train employees on emergency procedures. This can help in minimising downtime and losses.
Efficient loading and unloading protocols	Train staff on efficient loading and unloading procedures to minimise the duration for which the doors remain open. By implementing quick turnaround practices, one can significantly reduce the risk of temperature fluctuations.
Thermal imaging	Use of thermal imaging technology to identify gaps and leakages within the facility which are usually difficult to identify with the naked eye.
Roles and responsibilities of personnel involved	Clearly define the duties and responsibilities of each individual involved in the maintenance and operation of the facility, ensuring accountability and smooth operations.
Maintenance and monitoring including preparation of maintenance schedule	Conduct regular inspections and maintain equipment and facilities in accordance with a predetermined maintenance schedule.
Preparation of calibration schedule for equipment	Establish a schedule for calibrating temperature control equipment to maintain accuracy and reliability.
Adoption of standardized / best practices for temperature validation methods	Implement proven methods and protocols for validating temperature control, ensuring product integrity and regulatory compliance throughout the storage process.
Decommissioning of old systems	Regular assessment, prioritising replacement of old equipment, modernisation, training and transition plans, proper disposal, documentation, and continuous improvement.

2.3. Reefer Transport

2.3.1. Introduction

Refrigerated containers, commonly known as reefers, constitute an integral component of the cold chain, linking the various segments and ensuring the continuity of optimal conditions for perishables and vaccines throughout the supply chain, thereby preserving their quality. Reefers transport temperature-sensitive goods such as fruits, vegetables, pharmaceuticals, processed food, and meat. These containers are powered by generators that supply electricity.

Reefer vehicles are pivotal within the cold value chain for several reasons:

Preserve Freshness and Reduce Wastage. These vehicles are fitted with refrigeration units that regulate and maintain temperatures within the required ranges, effectively preserving the freshness of the cargo. In addition to maintaining freshness, they also inhibit spoilage by establishing a controlled environment that retards degradation processes, including bacterial proliferation and chemical reactions.

Table 7: Reefer classification [18]

Classification	Length categorisation
Small and battery powered pickup trucks	<8 ft
Vehicle powered units	<14 ft
Self-powered units	<30 ft
Trailer units	>30 ft

Figure 7: Reefer vehicle and its interior [19]

2.3.2. Transitioning to sustainable options

Transitioning to sustainable options may be realised through two principal methods: retrofitting existing structures or implementing new setups from scratch. Retrofitting entails the enhancement of current buildings with energy-efficient technologies, the integration of renewable energy, and water conservation measures. This approach recognises the significance of maximising the lifespan of existing infrastructure whilst aligning it with modern sustainability standards. Conversely, establishing entirely new setups permits the incorporation of state-of-the-art green building designs, zero-energy principles, environmentally friendly materials, and integrated sustainable infrastructure. Whether upgrading existing structures or commencing anew, these strategies play an indispensable role in promoting environmental responsibility and diminishing overall ecological footprints.

2.3.2.1. Retrofitting

Below are the potential retrofits in the following areas:

1. Insulation
2. Cooling technology
3. Other design interventions

1. Insulation

Retrofit option 1: Upgrading from EPS, Mineral wool to PUF / PIR

Retrofitting from traditional materials such as EPS or mineral wool in reefer vehicles signifies a considerable enhancement in thermal performance and energy efficiency. PUF and PIR insulation exhibit superior performance that augments temperature regulation, which is vital for preserving the integrity of perishable goods during transportation. In addition to improved thermal performance, these materials are lightweight. Their minimal water absorption and robustness further guarantee long-term efficacy. For the flowchart, kindly refer to **Section 2.4.2.1 – Insulation.**

Cost analysis (*Illustrative example*)[20]:

The cost of PUF can vary depending on several factors like quality, thickness, location, and supplier. The average cost of PUF insulation can vary from INR 100-INR 250 per sq. ft.

For a size of 6 ft * 6 ft * 8 ft, the total PUF cost can vary roughly between INR 20k to INR 55k. Labour cost and other additional costs can sum this up to approximately INR 40k – INR 80k, or maybe even more.

Calculation of energy savings in reefer can vary significantly because reefer is a mobile unit and it can travel across various geographies and temperature zones, which in turn will decide the payback period.

However, if we assume that a reefer vehicle travels for 12 hours a day consuming 4 units (4kWh) per hour within the similar climatic zone, the total power consumption will be approximately 48 units per day.

If we assume 10% savings through this retrofit option, roughly 4.8 units per day can be saved, translating to cost savings of 4.8*10 (unit cost) = INR 48 per day translating to INR 18k (approximately) per year. In this case, the payback can be 2-4 years.

However, the payback can be as low as less than a year as well.

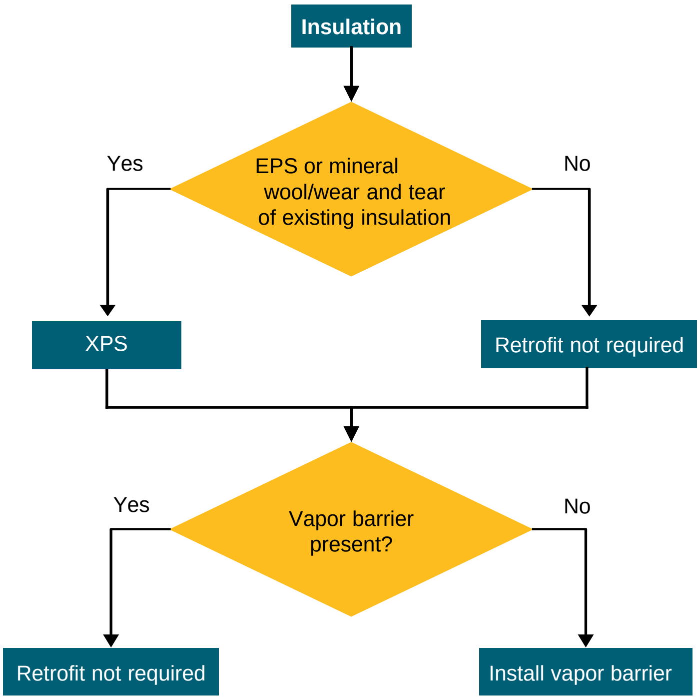
Please note that these are just indicative figures. For actual Return on Investment (RoI) and payback period, please consult with a reefer expert prior to initiating any retrofit.

Retrofit option 2: Upgrading from EPS, Mineral wool to XPS

For reefer vehicles, XPS can also be a viable option. XPS has been extensively utilised in smaller refrigerated vehicles that serve rural areas, particularly for vaccines. This is due to the fact that, for smaller vehicles, inert materials can be readily fabricated, and the insulation property remains consistent despite a reduction in size. XPS offers superior moisture resistance in comparison to PUF, as well as a higher R-value per inch, which results in enhanced insulation efficiency. Furthermore, it maintains its long-term R-value more effectively than PUF.

The long-term R-value of XPS does not diminish over time. The replacement of previous insulation with XPS is straightforward and convenient. In the retrofitting process of XPS, the panel remains intact, and only the XPS layer needs to be replaced. This contrasts with PUF, where the entire panel requires replacement.

Figure 8: Retrofitting of insulation in reefer



Cost analysis (*Illustrative example*)[21]:

The cost of XPS can vary depending on several factors like quality, thickness, location, and supplier. The average cost of XPS insulation can vary from INR 60–INR 200 per sq. ft.

For a size of 6 ft * 6 ft * 8 ft, the total XPS cost can vary roughly between INR 15k to INR 50k. Labour cost and other additional costs can sum this up to approximately INR 30k – INR 80k, or maybe even more.

Calculation of energy savings in reefer can vary significantly because reefer is a mobile unit, and it can travel across various geographies and temperature zones, which in turn will decide the payback period.

However, if we assume that a reefer vehicle travels for 12 hours a day consuming 4 units (4kWh) per hour within the similar climatic zone, the total power consumption will be approximately 48 units per day.

If we assume 10% savings through this retrofit option, roughly 4.8 units per day can be saved, translating to cost savings of 4.8*10 (unit cost) = INR 48 per day translating to INR 18k (approximately) per year. In this case, the payback can be 1–4 years.

However, the payback can be as low as less than a year as well.

Please note that these are just indicative figures. For actual Return on Investment (RoI) and payback period, please consult with a reefer expert prior to initiating any retrofit.

2. Cooling technology

Retrofit option 1: Integrating VFDs for energy efficiency

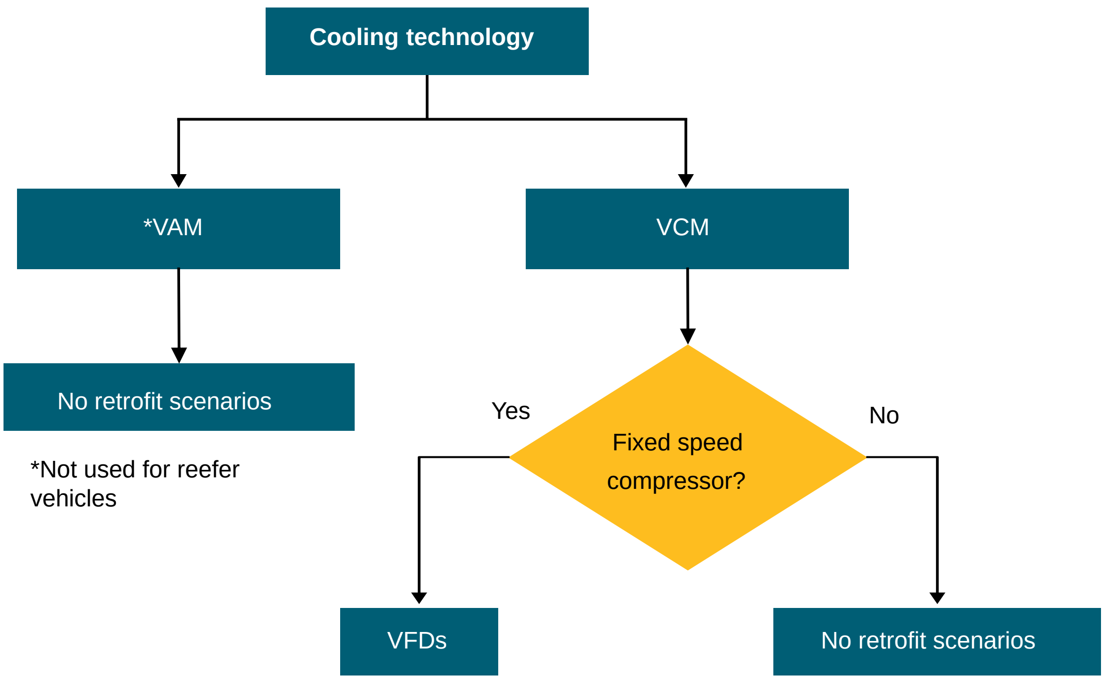
In the case of a reefer container, vapour compression is the traditional and most commonly utilised cooling technology. A potential retrofit option could involve the installation of VFDs in the event of fixed-speed compressors. This would aid in reducing energy consumption when the load is less.

The cooling load within a reefer can fluctuate considerably during transit, influenced by factors such as external temperature, cargo type, and loading/unloading activities. VFDs in reefers ought to dynamically adjust to these varying loads.

Given that reefers are mobile and operate under diverse environmental conditions throughout transportation, VFDs designed for such units may require additional features to cope with vibrations, shocks, and fluctuating temperatures.

The integration of VFDs into the overall control system of a reefer could present increased complexity due to the unit's mobile nature. Communication protocols and interfaces might necessitate customisation to meet specific transportation requirements.

Figure 9: Retrofitting - VFD



Please refer to **section 2.4.2.1 – Cooling technology** for more details.

It is advised to consult a refrigeration expert or a consultant for proper guidance and calculations.

Retrofit option 2: Integrating higher efficiency motors for energy efficiency

Please refer to **section 2.4.2.1 – Cooling technology** for more details.

3. Other Design Interventions

Retrofit option 1: Smart monitoring system for a reefer vehicle

Temperature monitoring is an essential aspect of reefer operations, playing a pivotal role in maintaining the integrity of the cargo during transport. It ensures that the quality of the goods is preserved, thereby preventing spoilage or damage. Furthermore, the sector is governed by stringent regulations mandating the installation of temperature monitoring systems to adhere to compliance standards. While these regulations are particularly rigorous in Western countries, it is anticipated that, with India's rapid advancement in the cold chain sector, similar compliance requirements will soon become relevant. Therefore, smart monitoring will play an important part in the future to meet these regulations.

Cost estimate[22]:

The approximate total cost of installation of a smart cloud-based temperature monitoring system can range between **INR 25K – INR 1 lakh**. It can go even higher depending on the container size, and the vendor providing the service. This includes the one-time hardware cost, subscription-based cloud software cost, installation cost and the services and maintenance costs.

Assuming that the smart monitoring system helps to preserve energy and reduce wastage resulting in average cost savings of **15-20%** on an average, the payback can be assumed between **3-6 years**. The energy saving cost will depend on various factors:

- Nature of cargo
- Ambient temperature and temperature to be maintained
- Type of insulation used and its wear and tear condition

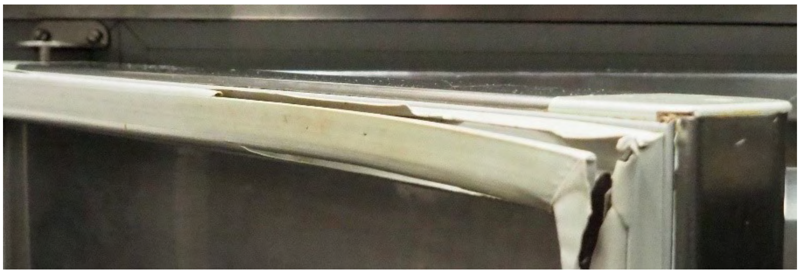
Please note that these are just indicative figures. For actual RoI and payback, please consult with a reefer expert prior to initiating any retrofit.

Retrofit option 2: Replacing worn-out ineffective gaskets & seals

An effective strategy for optimising the cold chain in reefer vehicles involves retrofitting by replacing worn-out and ineffective gaskets and seals. Over time, these components may deteriorate, leading to increased heat infiltration and compromised temperature control within the refrigerated compartments. By undertaking this retrofit, reefer vehicles can restore the integrity of their seals, preventing undesirable heat exchange and maintaining the desired low temperatures. This not only enhances the efficiency of the cold chain but also ensures the preservation of perishable goods during transit, aligning with the stringent requirements of temperature-sensitive cargo.

Cost analysis for this retrofit option can vary based on several factors, which cannot be factored in simultaneously. Therefore, it is advisable to consult a refrigeration expert to obtain details on the cost and payback.

Figure 10: Worn out gasket [23]



Retrofit option 3: Strengthen the insulation around the door to prevent warm air infiltration

An additional critical retrofitting option for reefer vehicles involves enhancing the insulation around the doors to prevent the infiltration of warm air. Frequently, inefficiencies in door insulation can result in temperature fluctuations, thereby compromising the overall efficacy of the cold chain. By reinforcing the insulation, reefer vehicles can minimise heat exchange, avert the infiltration of warm air, and sustain a consistently low-temperature environment. This proactive measure not only safeguards the quality of the transported goods but also contributes to energy efficiency by diminishing the workload on the refrigeration system. In essence, these strategic retrofits play an indispensable role in ensuring the reliability and performance of reefer vehicles within cold chain logistics.

The cost analysis for this retrofit option can vary, contingent upon several factors that cannot be simultaneously considered. Consequently, it is recommended to consult a refrigeration expert to get details on the cost and payback period.

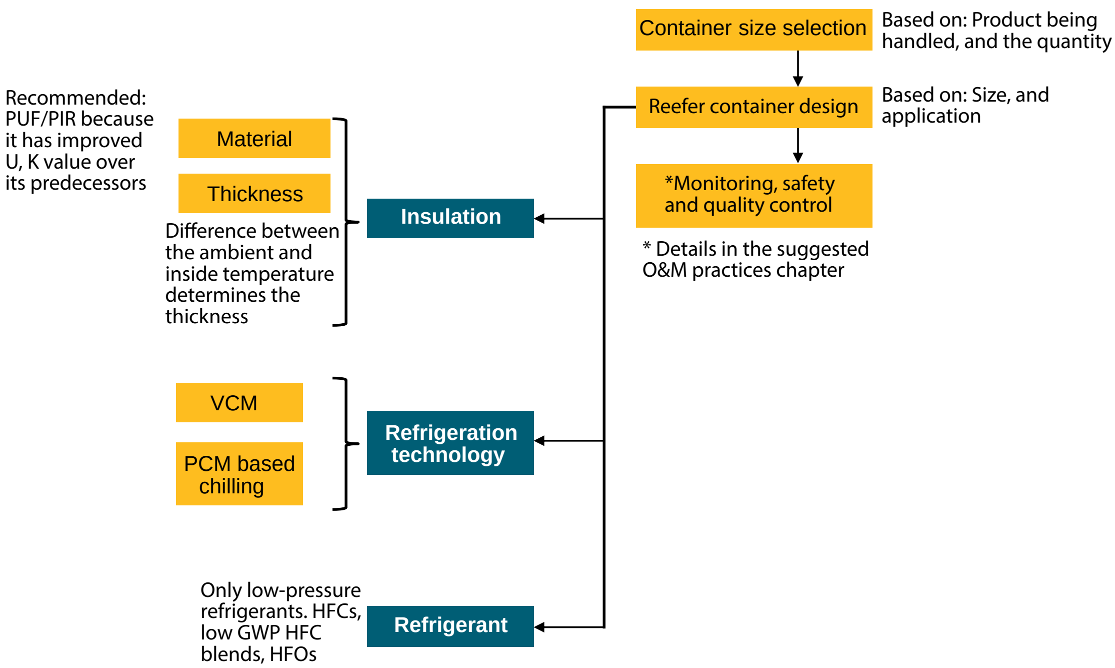
2.3.2.2. New Setup

In the refrigeration technology, apart from vapour compression, PCM (Phase Change Material)-based chilling is also being used by some reefer players.

The PCM-based technology utilises phase change materials to store and release thermal energy. These materials undergo a phase transition between solid and liquid states at a specific temperature, absorbing or releasing a significant amount of energy in the process.

PCM material selection is a very crucial part. It should be based on their phase transition temperature, which aligns with the desired temperature range for the refrigerated container. Common PCMs are paraffin wax, salt hydrates, and organic compounds. Higher thermal conductivity facilitates faster heat transfer between PCM and its surroundings, improving the overall efficiency. Other factors that need to be considered are latent heat properties, compatibility with the application, and integration with the system design.

Figure 11: New setup for a reefer vehicle



There is a small vapour compression unit present in the reefer vehicle to charge the PCMs. This vapor compression unit mostly uses freons like R-134a to charge the PCM.

In consideration of refrigerants, solely low-pressure variants are permissible for utilisation within reefer vehicles to circumvent any form of risk throughout the transportation process.

It is advised to consult a refrigeration expert prior to initiating any kind of actions for setting up a new reefer vehicle.

Figure 12: PCM based reefer [24]

PCM-based reefer vehicles are ideal for intercity transportation and short-haul services as their holding time is 14-16 [25] hours. PCM offers up to 80% [26] savings in operation costs by eliminating the use of diesel to run the refrigeration system.

> **Note:** *Reciprocating compressors have long been employed in reefer containers. However, scroll compressors are increasingly utilised in recent times owing to their ease of maintenance. As there are fewer moving parts, the maintenance required is reduced.*

2.3.3. Suggested O&M practices

Table 8: Suggested O&M practices for reefer vehicles

Segment	Suggested Practices
Pre-loading inspection	Conduct a thorough inspection to check the overall condition of the vehicle, including tires, lights, and refrigeration unit. Verify that the reefer unit is clean and free from any debris or obstructions.

Segment	Suggested Practices
Loading and unloading procedures	Ensure proper loading and unloading procedures to maintain an even air distribution within the cargo area. Minimise the duration for which the doors remain open during loading and unloading process to avert temperature fluctuations.
Fuel management	Regularly inspect and maintain the fuel system to ensure a consistent and adequate fuel supply for the refrigeration unit. Monitor fuel consumption and efficiency.
Insulation checks	Inspect the insulation of the cargo area for any damage or wear. Repair or replace insulation as needed to maintain thermal efficiency.
Driver training	Train drivers on proper reefer unit operation, including temperature control, defrosting procedures, and emergency protocols. Emphasise the importance of careful driving to minimise shocks to the cargo.
Emergency preparedness	Equip the vehicle with emergency equipment, such as backup power sources, temperature recording devices, and tools for basic repairs. Train personnel on emergency response procedures in case of refrigeration unit failures or temperature excursions.
Documentation and compliance	Maintain appropriate temperature and RH monitoring logs along with any other documentation required for compliance.
Roles and responsibilities of personnel involved including loader, unloader, and other crew members shall be defined	Clearly define the duties and obligations of each individual involved in the process, ensuring accountability and smooth operations.
Labelling and marking of packages and reefer trucks	Implement a comprehensive labelling and marking system for both packages and reefer trucks to facilitate easy identification, tracking, and handling of goods.
Storage, lashing and locking practices	Establish proper procedures for the storage of goods within reefer trucks, including securing them with appropriate lashing and locking mechanisms to prevent shifting or damage during transportation.

Segment	Suggested Practices
Maintenance and monitoring including preparation of maintenance schedules	Regularly inspect and upkeep the equipment and system, adhering to a pre-defined maintenance schedule to prevent any breakdowns.
Training for other members like loaders, unloaders, and other crew members	Provide comprehensive training programmes to all the personnel involved in the process, including loaders, unloaders, and other crew members to ensure that they are equipped with the necessary skills and knowledge to perform their duties efficiently.
Decommissioning of old systems	Regular assessment, prioritisation of replacement of old equipment, modernisation, training and transition plans, proper disposal, documentation, and continuous improvement.
Troubleshooting of reefer vehicles	Establish a comprehensive protocol for identifying, analysing, and rectifying potential issues encountered with reefer vehicles, encompassing challenges such as temperature variability, mechanical breakdowns, and electrical irregularities.

2.4. Cold Storages

2.4.1. Introduction

Cold storage facilities are essential elements within the cold supply chain, playing a crucial role in maintaining the quality and prolonging the shelf life of perishable items. These units are kept at low temperatures, in accordance with product specifications, to protect perishables like fruits, vegetables, dairy products, and pharmaceuticals from spoilage. They help to mitigate post-harvest losses and enable the year-round availability of seasonal produce. Ranging from expansive warehouses to small refrigerated containers, cold storage solutions are designed to cater to the varied requirements of different industries, ensuring the uninterrupted distribution and delivery of temperature-sensitive commodities throughout the region.

Figure 13: Cold storage [27] [28]

2.4.2. Transitioning to sustainable options

Transitioning to sustainable options in cold storages is essential for mitigating environmental impact and fostering long-term resilience in the face of climate challenge. This transition is important for maintaining a balance between meeting growing demands for refrigerated storage and minimising the environmental impact.

2.4.2.1. Retrofitting

Below are the potential retrofits in the following areas:

1. Insulation
2. Cooling technology
3. Refrigerant
4. RE Integration
5. Other design interventions

1. Insulation

Retrofit option: Traditional insulating materials like EPS, XPS, mineral wool to PUF/PIR

PUF and PIR are recognised as superior solutions for insulation, outperforming EPS, XPS, and mineral wool due to their exceptional insulative properties. PUF offers a range of versatile installation methods, including the application of sandwiched panels and the use of PU foam spray on walls. The former technique entails the direct affixation of PUF sandwiched panels to walls and roofs, whereas floors are typically insulated using XPS or EPS panels.

The subsequent step involves verifying the presence of a vapour barrier atop the insulation. Should a vapour barrier be absent, it is advisable to install a layer of vapour barrier over the insulation. This measure will inhibit vapour from condensing around the insulation, thereby enabling the insulation to fulfil its intended efficacy.

Figure 14: Retrofitting of insulation in cold storages

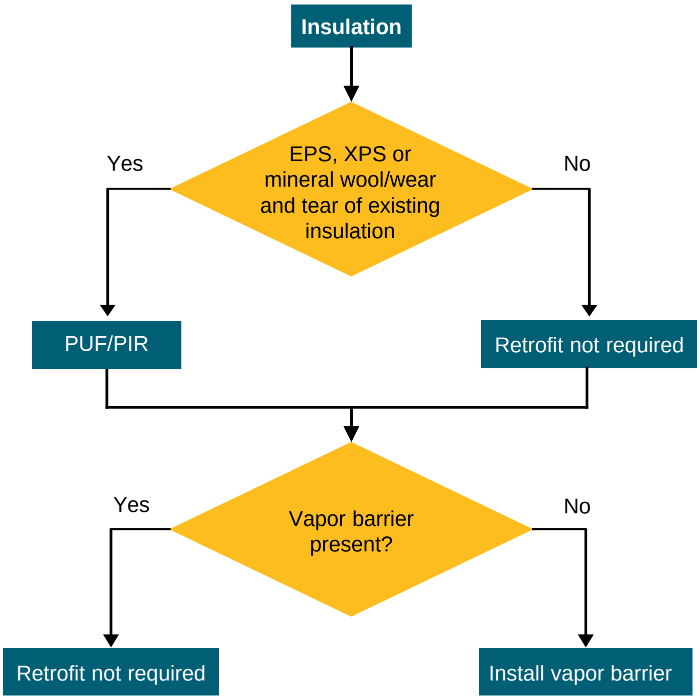


Table 9: PUF specifications with cost [29]

PUF panel thickness	Approximate temperature range (oC)	Cost range per sqmt (INR)
60mm	0 to +8	1400 – 1600
80mm	-5 to 0	1600 – 1800
100mm	-10 to -5	1800 - 2000
120mm	-25 to -10	2000 - 2300
150mm	-35 to -25	2300 - 2900

Cost estimate (*Illustrative example*)[30]:

For a 6 ft *6 ft * 8 ft cold room, the total cost of retrofitting to PUF insulation would be **approximately INR 50,000 for a 60mm thickness.** This cost is including all the labour costs.

The approximate **time duration required would be 2-3 weeks**, which again depends on the location and access to the facility.

Determining the payback period can be difficult for insulation because it depends upon several factors like energy savings observed, maintenance practices and maintenance costs, facility's usage pattern, temperature requirements, etc.

It is advised to consult a refrigeration expert for economic feasibility prior to initiating any retrofit.

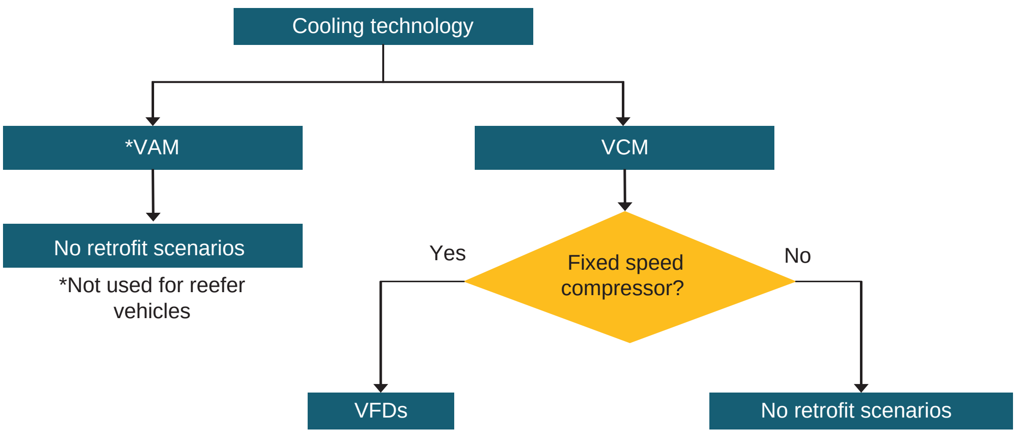
Note: *Rice husk has traditionally been used as insulation in Northern India, with facility owners praising its performance and ease of retrofitting. Given its effectiveness and simplicity in application, it presents a viable option for retrofit projects.*

2. Cooling technology

Retrofit option 1: Adding a VFD to the compressor

In the context of retrofitting compressors within a cold chain facility, the incorporation of inverter technology through VFDs proves to be exceedingly beneficial. This is particularly advantageous regarding the energy savings accrued in scenarios where the facility undergoes variable loads or experiences substantial fluctuations in external climatic conditions over the course of the year.

Figure 15: Retrofitting of VFDs



Cost estimate[31]:

A complete VFD setup for a **300TR** capacity facility would cost approximately **INR 5 lakhs**. The VFD would be attached to the compressor motor and the fan motors of the evaporator and condenser. The compressor will have the largest VFD of all in usual cases.

VFDs can result in energy savings of 15-20%. Depending on the case, with load variations, it can even reach 40%.

(Illustrative example)

Assume that there is a 60hp motor that runs for 15 hours a day and 300 days a year. Unit cost of electricity can be assumed as INR 8.

OPEX savings

The operational cost = power (kW) * running time (hours) * unit cost

Cost without VFD = 60 (hp) * 4500 * 8 = **INR 16,10,712**

Assuming VFD runs the motor at 100% speed for 30% of the time, 75% speed for 55% of the time and at 50% speed for the remaining 15% of the time. With this, the total cost would be:

Cost with VFD = 60 (hp) * 1350 * 8 + 25 (hp) * 2475 * 8 + 7.5 (hp) * 675 * 8 = **INR 8,68,878**

Total cost difference **= INR 7,41,834 = Cost savings per year**

Please note that these are just indicative figures. For actual RoI and payback, please consult with a cold chain expert prior to initiating any retrofit.

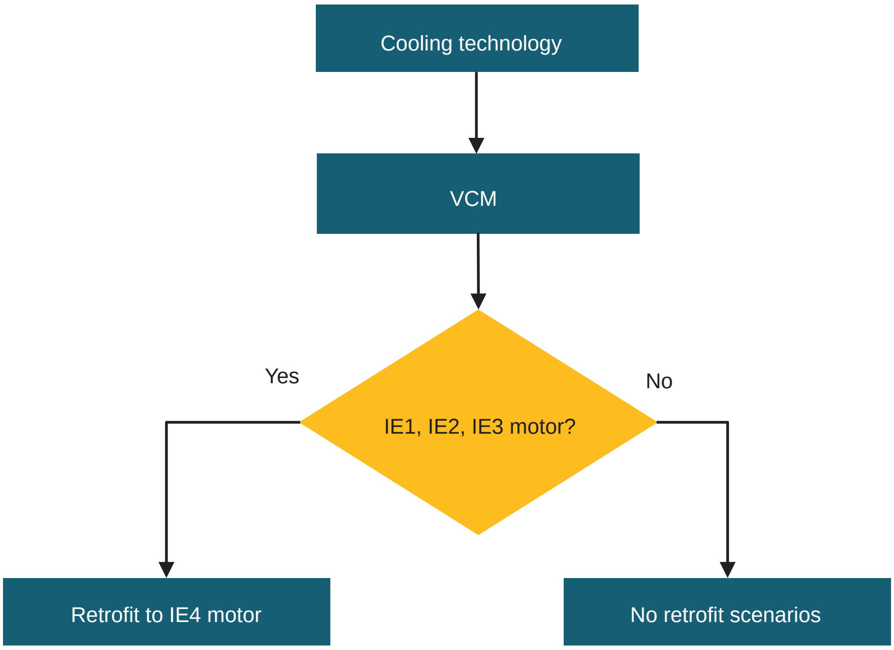
Retrofit option 2: Higher efficiency compressor motor

The IE stands for International Efficiency. 1, 2, 3 and 4 are the efficiency classes, which are described in Table 10. The IE classes were first developed in the year 2008, named IE 60034-30:2008. These classes were later revised in the year 2014, named IE 60034-30:2014.

Table 10: Efficiency class of motors [32]

Motor	Efficiency class	Efficiency for a 4-pole motor
IE1	Standard efficiency	77.2% approx.
IE2	High efficiency	82.8% approx.
IE3	Premium efficiency	85.3% approx.
IE4	Super premium efficiency	>90%

Figure 16: Retrofitting of higher efficiency class motors



The upcoming edition of IEC 60034-30-1 will include the IE5 efficiency class, aiming for a 20% reduction in energy losses compared to IE4. However, it's important to note that the IE5 standard is still under development and is not yet available in the commercial market.

It is usually recommended to retrofit to a higher efficiency class motor due to higher efficiency, resulting in energy savings. However, this is not always the case. It again depends on different scenarios. It could also be possible that 20-year-old motors which were manufactured with lots of copper are actually just as efficient as the new modern motors.

Cost Analysis: A lot of factors like size of the motor, existing motor conditions, running hours, cost of electricity etc. will contribute to the savings that can be potentially achieved for calculating payback period for this retrofit option.

Thus, it is suggested to consult with a cooling expert / consultant for individual retrofit scenario.

Retrofit option 3: Replacement of atmospheric condenser with evaporative condenser and stainless-steel coils

Figure 17: Evaporative condenser and water-cooled condenser[33]

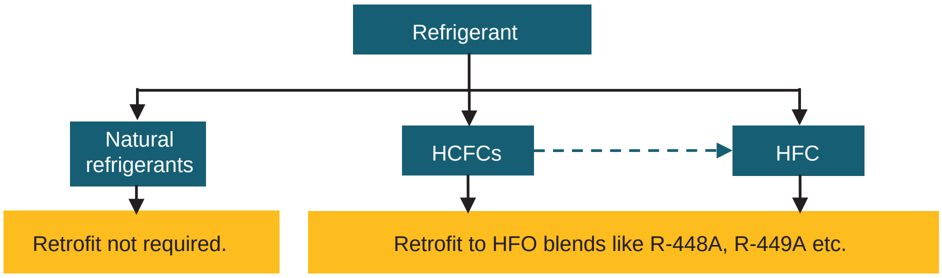
Evaporative condensers are generally more efficient than atmospheric condensers because they use water evaporation to remove heat from the refrigerant resulting in lower condensing temperatures and reduced energy consumption. This will result in lower drift losses, resulting in power savings of approximately 20-30%. Retrofitting in this scenario is not a very easy process because condensers are bulky and require a lot of layout and design planning prior to initiating the retrofit. It is therefore strongly recommended to consult a refrigeration and cold chain expert for retrofitting of the condenser unit.

3. Refrigerants

Natural refrigerants such as ammonia and carbon dioxide are environmental friendly because they possess negligible GWP and ODP values. Consequently, there is no necessity for retrofitting in such scenarios. Regarding Freons, which are chemical refrigerants, certain measures are being undertaken to mitigate their environmental impact. HCFCs are being phased out by the government under the HPMP Phase II. In due course, HFCs will also be eliminated. HCFCs such as R22 are scheduled for phase-out in the forthcoming years. This necessitates retrofitting to alternative refrigerants to facilitate the phase-out process. One immediate short-term retrofit solution could be the substitution of HCFCs with HFCs and HFC blends. Although HFCs and HFC blends have GWP values in a range similar to that of HCFCs, they offer comparable performance and efficiency, rendering them an appropriate retrofit option for the immediate future. In the long term, either HFOs and their blends may be considered, or natural refrigerants, which are likely to be the superior choice.

It is crucial to note that ammonia is a toxic refrigerant and must be handled with utmost care in order to prevent any leakages within the facility. Furthermore, it is recommended that any cold chain facility utilising ammonia as a refrigerant be situated outside urban limits for safety reasons, or it should implement stringent safety protocols. The staff must be properly trained in the management of ammonia, including regular safety checks to ensure a secure environment.

Figure 18: Retrofitting of refrigerants



> **Note:** *Some members of the European Union are imposing ban on certain HFO and HFO blend refrigerants as PFAS substances are known to be persistent in the environment, contaminating groundwater, surface water and soil, and causing serious health effects. However, no further developments have been reported on this matter.*

Retrofit option 1: R-22 to R-404A

Owing to the fact that R-22 will be phased out soon, this is a possible retrofit option. However, facilities using R-22 can continue using it.

The cost per kg for R-404A varies between INR 500 – 800. Here we have assumed the cost to be INR 800 for the calculations.

Cost estimate:

Table 11: Retrofitting from R-22 to R-404A[34]

Calculation for a 300 TR capacity facility			
Refrigerant	Cost per kg (INR)	Qty required (kg)	Cost of refrigerant (INR)
R-404A	800	250	2,00,000
Oil	Cost per litre (INR)	Qty required (Litre)	Total cost of oil replacement (INR)
POE oil	900	70	63,000

Calculation for a 300 TR capacity facility	
Nitrogen cylinders required for flushing up the entire system	5
Cost of 1 cylinder (INR)	12,000
Total cost of nitrogen flushing (INR)	60,000
Expansion valve (thermostatic)	12,000
Total cost of retrofitting from R-22 to R-404A (INR)	3,35,000 + labour charges

Above is an approximate costing and may vary from vendor to vendor.

Note: *There is an approximate of 5-7% drop in the refrigeration capacity after retrofitting from R-22 to R-404A, however, R-404A gives a higher system efficiency as compared to R-22.*

The calculations and the cost may vary based on various factors and therefore it is advised to consult a refrigeration expert prior to initiating any retrofit.

Retrofit option 2: R-22 to R-407C

The cost of R-407C varies from INR 400 to INR 800. Here we have assumed the cost to be INR 500 for the calculations.

Table 12: Retrofitting from R-22 to R-407C [35]

Calculation for a 300 TR capacity facility			
Refrigerant	Cost per kg (INR)	Qty required (kg)	Cost of refrigerant (INR)
R-407C	500	250	1,25,000
Oil	Cost per litre (INR)	Qty required (Litre)	Total cost of oil replacement (INR)
POE oil	900	70	63,000
Nitrogen cylinders required for flushing up the entire system			5
Cost of 1 cylinder (INR)			12,000
Total cost of nitrogen flushing (INR)			60,000
Expansion valve (thermostatic)			12,000
Total cost of retrofitting from R-22 to R-404A (INR)			2,60,000 + labour charges

Above is an approximate costing and may vary from vendor to vendor.

The calculations and the cost may vary based on various factors and therefore it is advised to consult a refrigeration expert prior to initiating any retrofit.

> **Note:** *R404A & R-407C will eventually be phased out under the Kigali amendment because they are high GWP refrigerants. The provided retrofit options are just temporary solutions.*

Retrofit option 3: R-22/R-404A/R-134A to HFOs and HFO blends

HFOs are increasingly recognised as long-term substitutes for current refrigerants owing to their low GWP, which mitigates their impact on climate change. Additionally, HFOs have a zero ODP. Nevertheless, a significant issue with HFOs is their cost, which is approximately three times higher than that of existing refrigerants on the market. This price disparity can be attributed to the absence of domestic manufacturing in India, necessitating imports. Consequently, there is a development of HFO blend refrigerants, such as R-448A and R-449A, which exhibit a lower GWP compared to HCFCs and HFCs, albeit higher than that of pure HFOs. With the expanding adoption of HFOs within India, it is anticipated that the cold chain sector will progressively transition to these refrigerants.

The uptake of HFOs in India is also experiencing a gradual progression, impeded by the knowledge gap among technicians who operate cold chain facilities.

4. RE integration

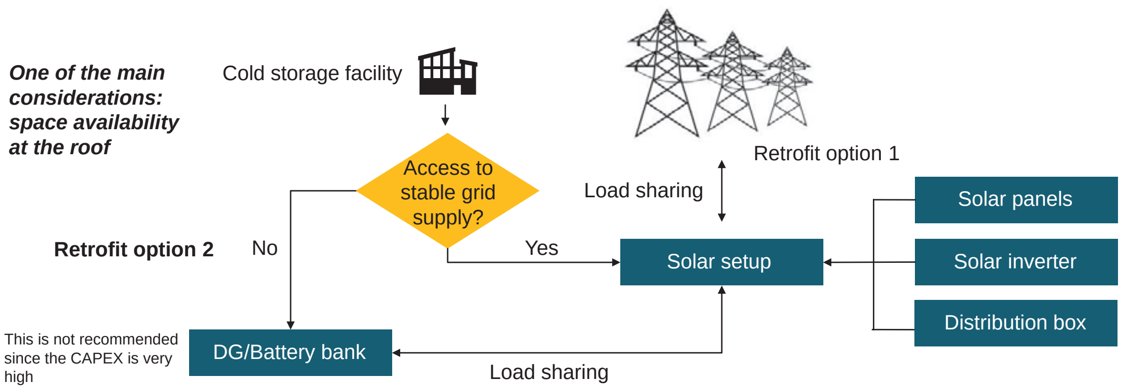
When considering RE options for integration within the cold chain segment, the selection process revolves around crucial factors such as the levelised cost of energy, land and water footprint, and O&M costs and challenges. The primary emphasis is on solar integration, where advancements in technology and established frameworks render it a viable and efficient choice. However, other RE technologies like geothermal and biomass hold promise but are currently at a very nascent stage. Development for seamless integration within the cold chain is under progress, with a few pilot projects being executed. These alternatives are being explored, taking into account their potential benefits, and thus our current focus remains on optimising solar solutions to ensure sustainability, cost-effectiveness, and minimal environmental impact in cold chain operations wherever feasible.

Solar Integration:

Prerequisite for solar integration:

- Availability of an electricity grid with constant and a stable electricity supply.
- The recommended solar capacity is 100% of the refrigeration load (7am – 5pm)

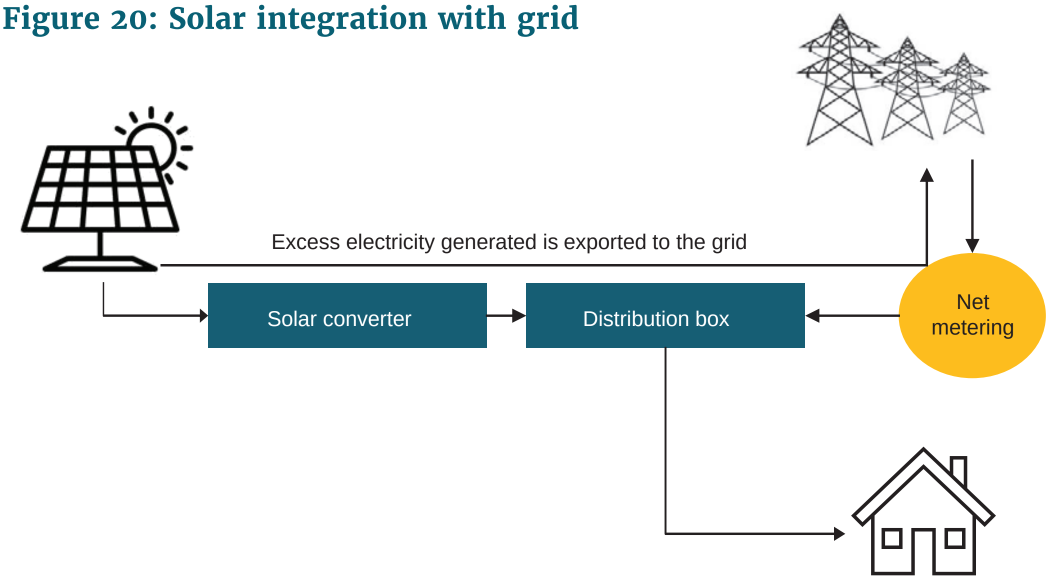
Figure 19: Solar integration in a cold storage



Retrofit Option 1 – Solar integration with electricity grid:

The solar integration can be done easily to any cold storage, provided that there is availability of electricity grid for load sharing. Based on the availability of area, solar panels can be installed for power generation.

Figure 20: Solar integration with grid



In this case, the solar electricity produced is consumed on site and only the differential electricity is consumed from the grid. In case of excess generation, the extra electricity units can be exported to the grid and consumed later on through the concept of net metering (regulations for net metering vary from state to state).

A solar setup for retrofitting into an existing facility primarily consists of solar panels installed on the rooftop of the cold storage facility, a solar inverter (to convert the DC current produced by the solar panels into AC current), and a distribution box.

Cost estimate:

Table 13: Cost estimation for solar integration with grid[36]

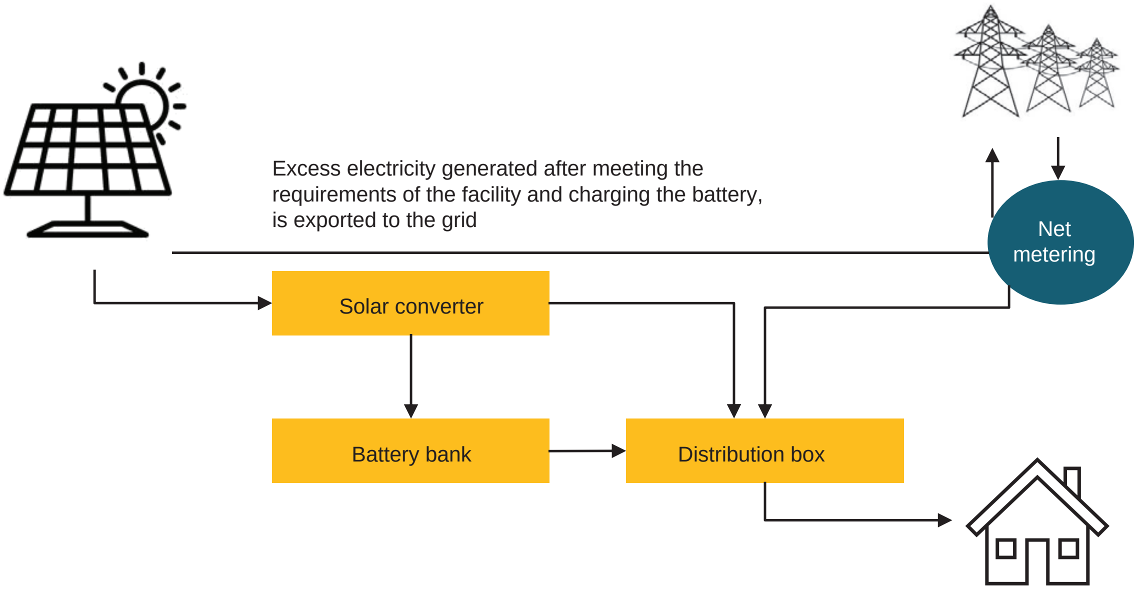
Parameter	Details	Assumptions
Area of solar panel	900 sq. mt. (75kW)	10-12 sq. mt. / Kw
Cost per panel	INR 30,00,000	INR 40 per watt
Installation cost	INR 12,00,000	INR 16 per watt
Inverter cost	INR 6,75,000	INR 9 per watt
Avg. generation per day	300 kWh (300 units)	4 kWh (4 units)
Capital Expenditure (CAPEX)	INR 48,75,000 excluding transportation cost	
Estimated savings per year	INR 8,76,000	300*365*8
Return on Investment	18% (assuming best-case scenario without net metering)	
Payback	5-6 years (assuming best-case scenario without net metering)	

Please note that these are just indicative figures. For actual RoI and payback, please consult with a solar integration expert prior to initiating any retrofit.

Note: *Various factors like power cuts, power export policies for unused solar power etc. have to be considered while accounting for the total number of units generated by solar per year*

Retrofit Option 2 – Solar integration with Battery bank:

Figure 21: Solar integration with battery bank



Cost estimate

Table 14: Cost estimate for solar integration with grid and battery bank[37]

Parameter	Details
Capacity	75kW
Initial CAPEX	INR 48,75,000 + transportation cost
Battery system cost	INR 18,75,000 (INR 25 per watt)
Total CAPEX	INR 67,50,000
Estimated yearly savings	INR 8,76,000
RoI	14% (considering the best-case scenario)
Payback period	7-9 years

Please note that these are just indicative figures. For actual RoI and payback, please consult with a solar integration expert prior to initiating any retrofit.

Solar integration with DG:

Integrating solar power with a diesel generator set, as opposed to the electricity grid, can be a costly endeavour due to a variety of factors. The necessity for system redundancy, the incorporation of energy storage solutions to ensure a continuous power supply during periods of low solar generation, the requirement for additional inverters and control systems to synchronise diverse power sources, and the ongoing fuel costs for diesel generators all contribute to higher initial and operational expenses. Maintenance costs for diesel generators can be relatively high, and the overall complexity of managing both solar and diesel components adds to the total expense. Although grid connection infrastructure might pose challenges in some instances, the long-term cost-effectiveness of grid-connected systems is often superior, providing backup power and utilising the grid as a buffer during periods of low solar generation. The scale of the installation, location, and technological advancements also play crucial roles in determining the overall cost dynamics of solar-diesel hybrid systems.

Factors affecting solar integration cost:

1. *Roof type. Lower costs are incurred when roof is already layered with GI sheets when compared to RCC or asbestos sheets.*
2. *AC cable length.*
3. *Site location: Material movement is an important factor for cost and better the site location connectivity, lower the cost.*
4. *Building height. Installation cost is higher in taller buildings.*

Note: A new setup of solar integration would also require the same steps which are mentioned for retrofitting.

Solar integration can be a very costly affair initially, due to which many stakeholders are skeptical to integrate solar energy, even though the OPEX becomes significantly lower. There are several schemes offered by the government at both central and state level to promote the wide adoption of solar energy.

The following schemes are offered by the central government as well as by various state governments. Concerned stakeholders are requested to liaise with an RE integration expert for more detailed information on the policies and for the purpose of implementation.

Table 15: National level policies for solar integration[38]

Policy / scheme	Description
National Solar Mission (NSM), 2010, 2011, 2013, 2015, 2016	Managing offtake risk by signing long-term power purchase agreements (PPA) and providing payment security. Low tariffs for discoms through competitive bidding. Financial support for project developers. Policy certainty and continued government support for promotion of RE.
Introduction of Renewable Energy Certificates (REC), 2010	Creating mechanism to broaden the demand base
National Tariff Policy, 2016	Creating demand for RE by setting common national trajectory for RE's share in consumption.

Table 16: State level policies for solar integration[39]

State	Policy / scheme
Andhra Pradesh	Solar Power Policy 2012, 2015, 2018 Solar – Wind Hybrid policy 2018
Bihar	Policy for Promotion of New and Renewable Energy Sources 2011, 2017
Gujarat	Solar Power Policy 2011, 2015
Karnataka	Karnataka Renewable Energy Policy 2009, 2016 Solar Policy 2011, 2014
Madhya Pradesh	Policy for Implementation of Solar Power Based Projects 2012
Maharashtra	New Policy for Power Generation from Non-Conventional Sources 2008 Comprehensive Policy for Grid Connected Power Projects based in New and Renewable (Non-conventional) Energy Sources 2015
Punjab	New and Renewable Sources of Energy Policy 2012
Rajasthan	Rajasthan Solar Energy Policy 2011, 2012, 2019 Rajasthan Wind and Hybrid Energy Policy 2019

State	Policy / scheme
Tamil Nadu	Tamil Nadu Solar Policy 2012, 2019
Telangana	Telangana Solar Power Policy 2015, 2016
Uttar Pradesh	Solar Power Policy 2013, 2017

5. Other design interventions

Retrofit option 1: Dock shelters and dock levelers

Dock shelters and dock levelers can serve as a valuable retrofit option for cold storage facilities, addressing the critical need for temperature control, energy efficiency, and operational improvements. These enclosures, installed around loading docks, create a sealed environment that minimises the exchange of air between the cold storage area and the external surroundings. By preventing the infiltration of outside air, contributing to energy efficiency by reducing the workload on refrigeration.

There are different types of dock shelters and levelers available in the market.

Traditional swing-open trailer doors typically leave 1- to 2-inch hinge gaps along both sides of the trailer, which equate to a 2.5-square-foot hole in the wall at each dock opening[40].

Cost estimate[41]:

Depending on the region, the unsealed gaps can cost up to **INR 1 lakh per year** approximately.

Along with the sides, the bottom end also allows some amount of energy to escape if not sealed properly. These losses can result in another **INR 15k to 70k per year.**

A dock shelter price can range anywhere from **INR 50k to 1.5 lakhs per door**, depending on the **dimensions of the door, type of dock shelter and the ambient surrounding temperature.**

An **inflatable dock shelter** for door dimension of **3500 * 3750mm (H*W)** will cost around **INR 55k.**

Assuming the yearly losses due to unsealed gaps to be INR 1 lakh, the payback can still be less than 2 years, in this case.

Please note that these are just indicative figures. For actual Return on Investment (RoI) and payback period, please consult with an expert prior to initiating any retrofit.

Figure 22: Retractable, cushion and inflatable dock shelters [42]

Figure 23: Dock levellers [43]

Retrofit option 2: Installation of sensors and PLC (Programmable Logic Controllers)

Automation plays a pivotal role in the quest for energy efficiency, as it facilitates optimised operations that enable precise control over various measurement parameters, such as temperature and humidity. It further permits real-time monitoring and the development of an energy management system. In the long term, these capabilities assist in managing energy more effectively and efficiently.

Cost analysis[44]:

For a **5000MT** capacity cold storage, the cost would vary somewhere around INR **10 lakhs (through MIDH subsidies)**

An estimated **energy saving of 3-4%** can be expected for this option.

Calculating the payback will depend on factors like load variation throughout the year, electricity tariffs, running time, energy saved, difference between the ambient and maintained temperature etc.

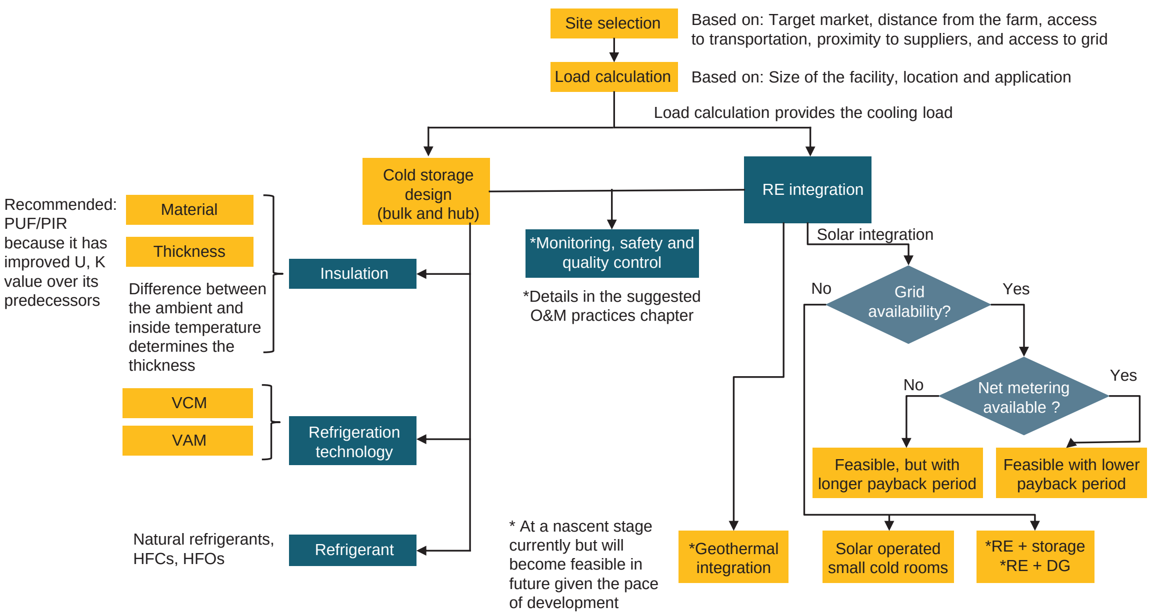
It is advised to consult a refrigeration expert or a consultant for proper guidance and calculations.

2.4.2.2. New setup

For a cold storage setup for perishables, the flowchart shown in *Figure 24* serves as an invaluable reference for decision-making in the establishment of a new system. Site selection is the initial criterion, which depends on several factors such as distance from the farm, access to transportation, target market, proximity to suppliers, and access to the grid. Following the heat load calculations (from which the cooling load is derived), the selection of the cooling technology, refrigerant, and insulating material takes place.

Among the natural refrigerants, CO_2 is another potential option, alongside ammonia. It is a high-pressure refrigerant. However, at present, it is not widely used. All the cold storage facilities have been utilising ammonia as the refrigerant for a considerable time, which has delivered proven results. CO_2 requires an entirely different system for operation. It is a relatively new refrigerant and, therefore, very few individuals possess expertise with CO_2 refrigerant. In the long run, it will eventually be accepted, owing to the fact that freons like HCFCs and HFCs are being phased out gradually and the dire need to switch to low ODP and GWP refrigerants.

Figure 24: New setup for a cold storage



Also, RE integration feasibility can be assessed based on the location and feasibility analysis. Post the design of the cold storage, monitoring, safety, and quality control measures should be defined, followed by documentation and compliance, overall ensuring an optimum functioning of the facility in terms of energy usage and sustainability.

Geothermal integration is an upcoming RE integration option with R&D currently taking place on it. In the near future, it would become a viable solution in the areas where natural hot springs are present. At present, this technology is being established in some areas of Himachal Pradesh like Kinnaur for apples.

For facilities that need to be established on leased lands, dismantlable structures can be developed, which are built on the structure of PUF panels and nuts and bolts. The building cost in such cases increase by approximately 10%.

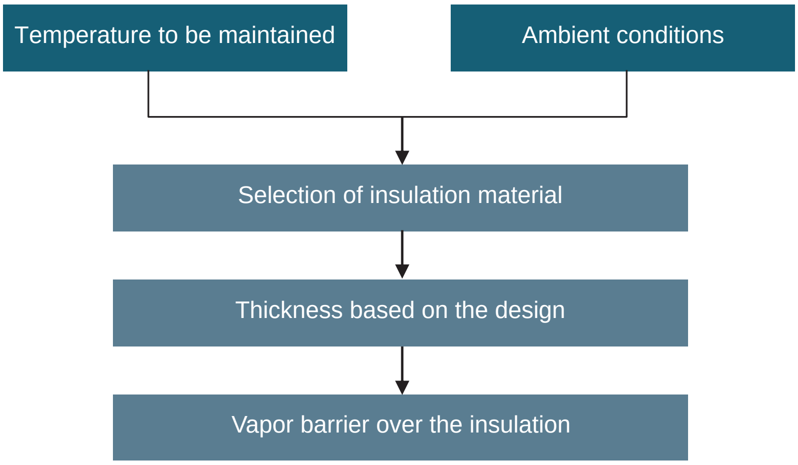
For solar integration, please refer to **section 2.4.2.1 – RE integration**.

It is advised to consult a refrigeration expert or a consultant for setting up of a new cold storage facility for proper guidance and calculations.

Insulation:

Figure 25: Cross section of Thermocol insulation and PUF

Figure 26: Insulation selection

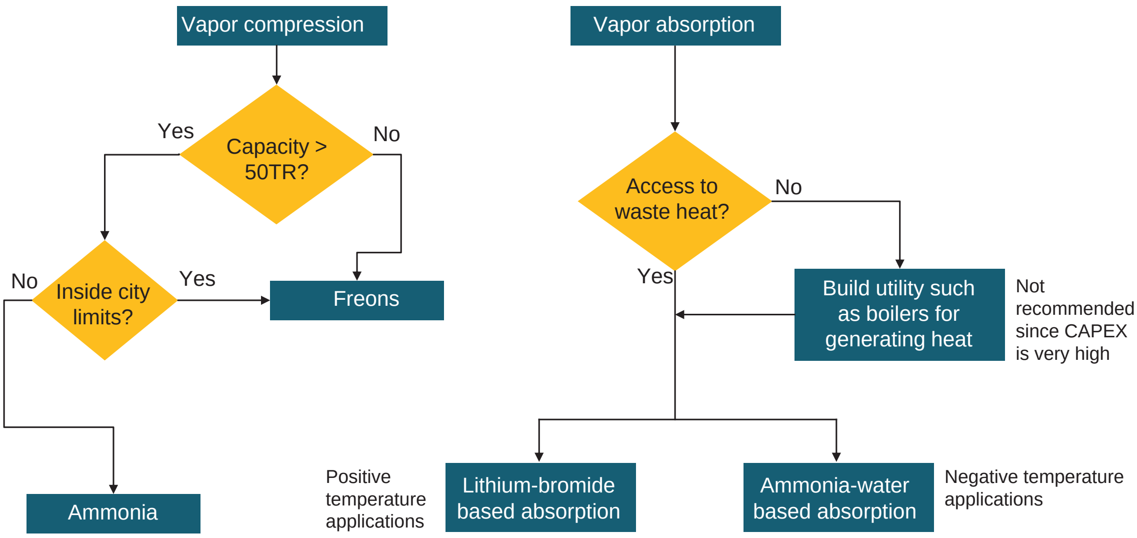


The primary insulation materials under consideration are PUF and PIR, with the determination of thickness guided by the temperature difference between the ambient and the facility.

In addition to the insulation, it is of the utmost importance to incorporate a vapour barrier over the selected insulation material. This serves as a crucial preventive measure, safeguarding against the formation of unwanted vapour within the facility.

Cooling technology:

Figure 27: VCM and VAM technology



Vapour compression (VCM): It is generally recommended to opt for ammonia-based system when the capacity is greater than 50 TR and location is outside the city limits because of the toxicity of ammonia. In any other case, freons can be opted for in a VCM technology.

Vapour absorption (VAM): As a cost-intensive technology, it is advisable to opt for it only when there is access to waste heat to drive the cooling mechanism. Should waste heat not be available, the CAPEX will rise due to the necessity of constructing heat-generating utilities.

Adsorption Refrigeration System (ARS): It is similar to VAM but operates on low pressure. It is a reliable off the grid solution and is being used for medium to large size cold storage facilities. They have custom boilers which do not require very high pressure. They work on waste heat generated usually from biomass.

Figure 28: ARS technology

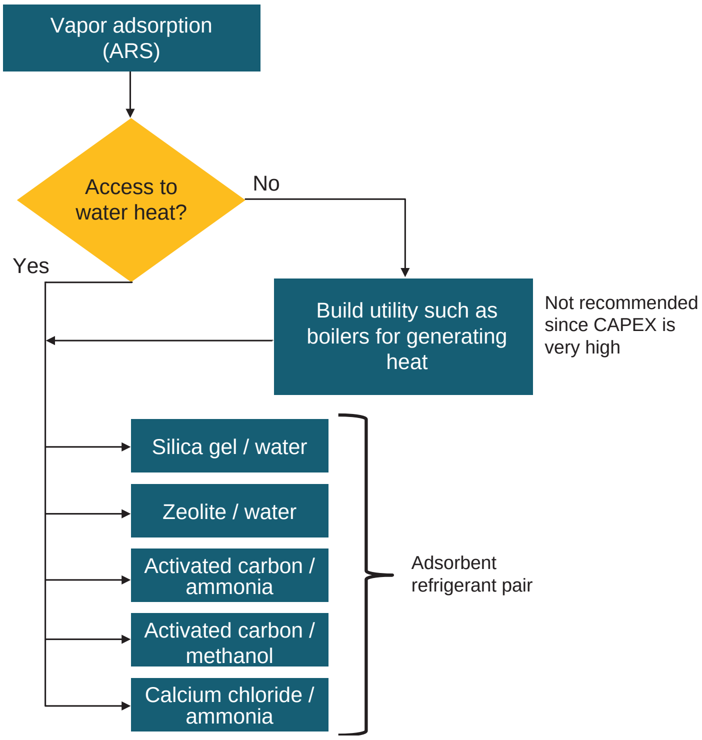
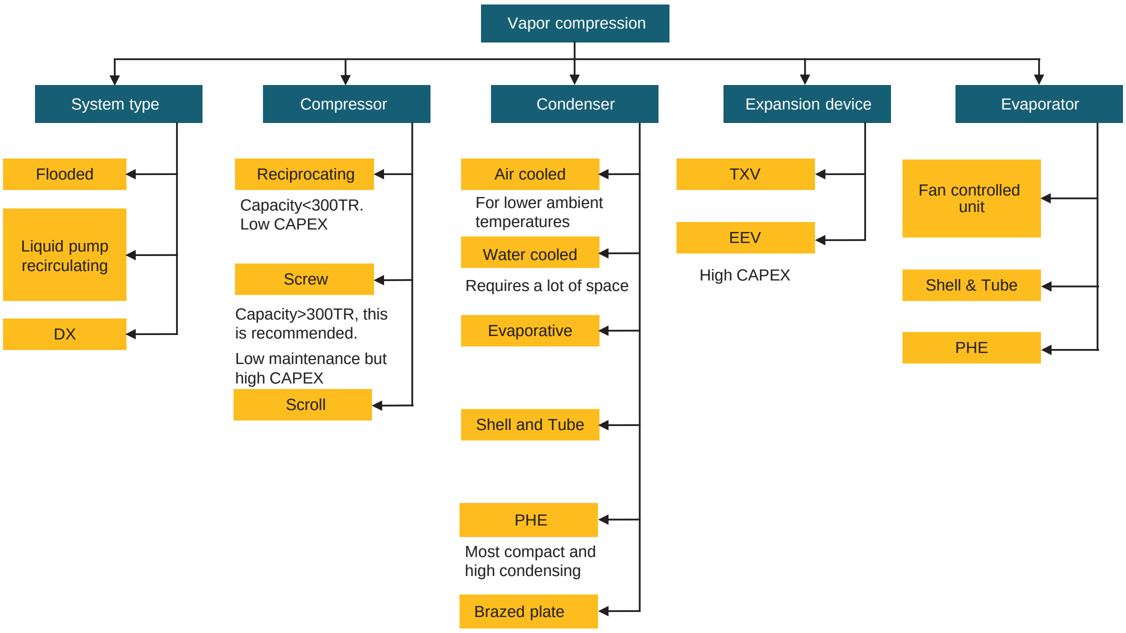


Figure 29: VCM system components



Refrigerants:

It depends on the refrigeration capacity needed.

In case the capacity is more than 50TR but the facility is located within city limits, then low charge ammonia system can be used, otherwise freons are recommended.

Ammonia is widely used as a refrigerant in India but since it is toxic in nature, it cannot be used as a refrigerant in facilities which are located or planning to be located within city limits without the necessary safety protocols and measures in place. This limitation has to be taken into account while setting up a new facility.

Even though it is a high-pressure refrigerant, its excellent thermodynamic properties and efficiency makes it one of the most favoured refrigerants.

In the case that the capacity of the facility is less than 50TR and the planned location is inside the city limits, it is recommended to go for freons as the refrigerant. In any other case, ammonia can be a good option.

HFOs are an alternative. They are ideal for use given their properties, however, due to the cost (because of limited production in India) it is not economically feasible as on date.

Carbon dioxide (CO_2) serves as a natural refrigerant with negligible ODP and GWP values, rendering it an excellent choice for such applications. Nevertheless, the issue with CO_2 lies in its high-pressure characteristics, which can render it hazardous if not managed correctly. Moreover, refrigerants under high pressure typically necessitate sturdy and intricate system components to cope with the elevated pressures. These considerations contribute to the lower adoption rate of CO_2 as a refrigerant both within India and internationally.

Other design interventions:

Ante rooms are a crucial design intervention that can be integrated into a new setup. They act as a buffer zone between the outside environment and the controlled temperature zones within the facility. They help to maintain the integrity of the cold storage environment by preventing warm air from entering when doors are

Figure 30: Ante room in a cold storage facility

open for loading and unloading. By serving as an intermediate space, ante rooms allow workers to enter and exit without significantly impacting the temperature inside the storage areas, this minimizing energy loss and ensuring consistent cold storage conditions. Additionally, ante rooms provide a space for workers to don appropriate protective gear, such as insulated clothing or gloves, before entering the cold rooms, helping to maintain their comfort and safety while working in low temperature environments.

2.4.3. Suggested O&M practices

Table 17: Suggested O&M practices for ammonia refrigerant

Segment	Suggested Practices
Leak detection	Ensure that if any leak occurs, it is detected well in time, before any serious hazard occurs. Cold storage facility owners should install ammonia leak detection sensors to promptly identify and address any leak.
Safety training	Provide thorough safety training to personnel working with or around the ammonia refrigeration system, including emergency response procedure and proper handling of ammonia.
Pressure testing	Periodically conduct pressure testing of system components to verify integrity and detect any weaknesses or vulnerabilities.

Table 18: Suggested O&M practices for cold storages

Segment	Suggested Practices
Administration	Implement and control storage activities effectively. Monitor and record temperature and humidity levels regularly. Implement and control a regular maintenance schedule for cold storage equipment. Maintain an inventory of spare parts for critical components.
Conducting operations	Optimise storage layout for efficient use of space. Implement a system for tracking inventory and shelf life.

Segment	Suggested Practices
Equipment status control	Regularly inspect and maintain refrigeration systems. Implement a monitoring system for cold storage equipment.
Operator Knowledge and Performance	Provide training on proper handling and storage procedures. Conduct regular drills for emergency response.
Work Control System	Conduct routine checks on insulation and door seals. Respond promptly to any temperature deviations.
Conduct of Maintenance	Address refrigerant leaks immediately and conduct regular leak testing. Clean and inspect evaporator and condenser coils regularly.
Preventive Maintenance	Schedule routine maintenance for compressors, condensers, and evaporators. Lubricate bearings and motors as per manufacturer recommendations.
Maintenance Procedures and Documentation	Develop clear procedures for routine maintenance tasks. Maintain comprehensive documentation for all maintenance activities.
Thermal imaging	Use of thermal imaging technology to identify gaps and leakages within the facility which are usually difficult to identify with the naked eye.
Decommissioning of old systems	Regular assessment, prioritisation of replacement of old equipment, modernisation, training and transition plans, proper disposal, documentation, and continuous improvement.

2.5. Ripening Chambers

2.5.1. Introduction

A ripening unit is an indispensable component of the cold chain sector, designed to expedite the ripening process in fruits and vegetables. Ripening chambers hold a central position within the cold chain sector, providing a regulated environment for the ideal ripening of fruits. These specialised chambers are equipped to control temperature, humidity, and ethylene levels, thus ensuring an exact and managed atmosphere that hastens the ripening process whilst preserving the quality and freshness of the produce. Utilising advanced technology and climate control systems, ripening chambers allow producers and distributors to oversee the ripening stages of fruits with great precision, thereby enhancing the overall efficiency of the supply chain. The introduction of ripening chambers is pivotal in prolonging the shelf life of fruits, diminishing post-harvest losses, and guaranteeing a consistent and high-quality supply of ripened produce to satisfy market demands.

Figure 31: Ripening chambers

Requirements for a ripening chamber are:

- Reasonably airtight room with insulation
- Temperature control system for cooling / heating
- Air circulation and ventilation system
- Humidity control
- Ethylene gas injection system
- Electric control system

There are mainly two types of ripening chambers:

Type 1:

In this type, the ripening chamber is equipped with a cold room and an ethylene generator. For maintaining the desired temperature and the humidity level, a ceiling mounted fin coil evaporator is connected to the condenser unit outside. Fruits and vegetables within the perforated plastic crates will be placed in the cold rooms and the cold air will be allowed to pass through these crates and uniform air circulation, uniform ethylene distribution, and ripening will be ensured. A ventilation system is provided to keep the CO_2 level within the desired limit.

Type 2:

These types of ripening chambers have a different type of air circulation system installed. This system helps to generate the desired static air pressure in the ripening chambers. The insulated cold rooms have a system of false ceiling, separated, and sealed annular space between wall and palletised crates / CFB boxes with or without air-inlet locking system to isolate designated pallets etc. Cool air is routed through false ceiling into boxes with perforated holes of Crates / CFB boxes for air circulation which, in turn are stacked in single / multi-tier system.

2.5.2. Transitioning to sustainable options

2.5.2.1. Retrofitting

Below are the potential retrofits in the following areas:

1. Insulation
2. Cooling technology
3. Refrigerant
4. RE Integration

1. Insulation

Retrofit option: EPS/XPS/Mineral wool to PUF/PIR

In the context of insulation choices for ripening facilities, where effective temperature control is paramount, PUF stands out as the predominant and highly recommended choice. PUF should have thickness of 60mm to 120mm and insulated sandwich panel of minimum density 40 kg/m^3. Floors shall have 60mm thick PUF sandwich panels and it is recommended to cover the floor insulation with 100mm concrete.[45]

Please refer to **2.4.2.1 – Insulation**, for more details.

2. Cooling technology

Please refer to **2.4.2.1 – Cooling technology**, for more details.

3. Refrigerants

According to the National Horticulture Board (NHB), only refrigerants with zero ODP and low GWP are recommended. These are typically either natural refrigerants or HFOs or their blends. However, if the current ripening facility utilises natural refrigerants such as ammonia, there would be no necessity to retrofit the refrigerants. It is crucial, however, to ensure that proper practices are adhered to, for prevention any leakages, given that ammonia is a toxic substance.

Conversely, in the instance of Freons, refrigerants with low GWP and zero ODP prove to be exceedingly costly in India, primarily due to the prerequisition of importing them. With the ongoing phase-out of HCFCs, it is essential to consider retrofitting to alternatives that can deliver comparable efficiency with minimal cost implications. Transitioning to HFCs and HFC blends represents a viable option for the immediate future.

Please refer to **2.4.2.1 – Refrigerant**, for more details.

4. RE integration

Please refer to **section 2.4.2.1 – RE integration** for more details on the aspects of solar integration.

> **Note:** *Ripening chambers often feature limited space, making the integration of solar energy into the facility challenging. However, if the ripening unit is situated within a cold storage facility, there emerges a viable opportunity to incorporate solar technology into the ripening chambers as well.*

2.5.2.2. New Setup

In recent years, there has been a marked increase in the establishment of modern ripening chambers in India, reflecting the heightened awareness of their importance within the agricultural and horticultural sectors. The demand for ripening chambers is rising owing to several factors such as the reduction of post-harvest losses, improvement of the ripening process, and enhancement of the overall market value of the produce.

Figure 32 provides a flowchart for step-by-step approach for new setup for a ripening unit.

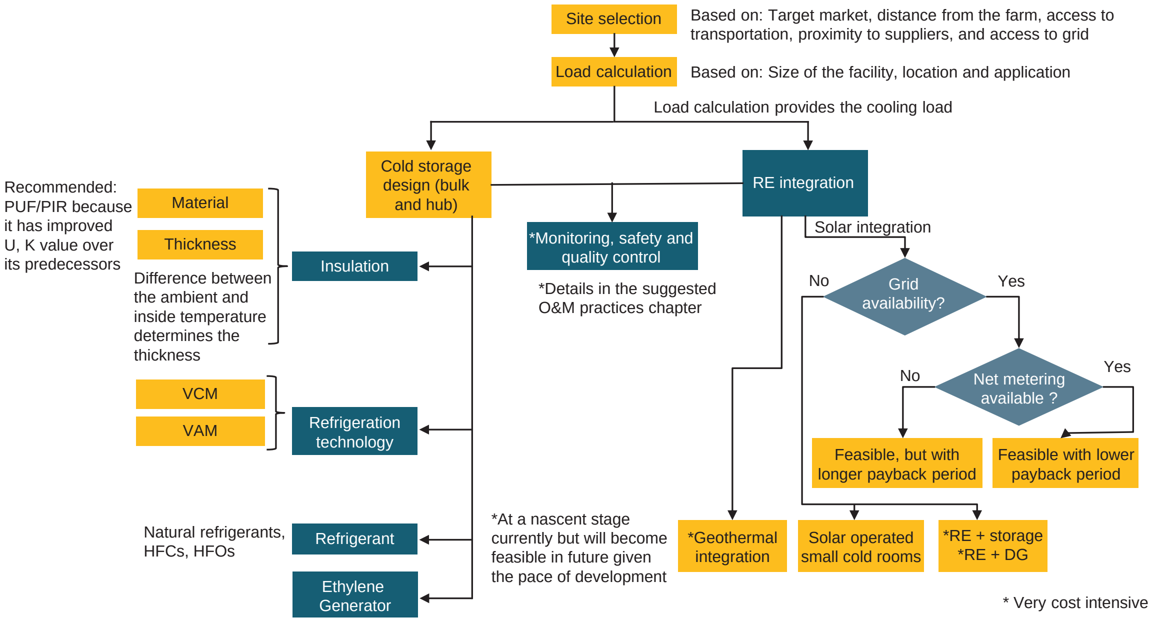


Figure 32: New setup for a ripening chamber

Refer to **section 2.4.2.2.** for more details regarding usage of CO_2 as a refrigerant.

An additional component apart from the regular ones discussed in the previous chapters, is ethylene generation. Ethylene plays a crucial role in the ripening process of fruits. In ripening chambers, ethylene is introduced into the chamber through ethylene generators. These generators release controlled amounts of gas into the chamber, allowing for precise regulation of the ripening process.

Ethylene can be introduced in the following ways:

- By using independent ethylene generator with cylinder, ethylene cartridges and ethylene-nitrogen mixture (5% ethylene + 95% nitrogen) cylinder

- Centralised ethylene supply with automation for multiple chambers for controlled and safe dosing of ethylene (for larger units)

It is advised to consult a refrigeration expert / consultant for setting up of a new ripening chamber facility for proper guidance and calculations.

2.5.3. Suggested O&M practices

Table 19: Suggested O&M practices for ripening chambers

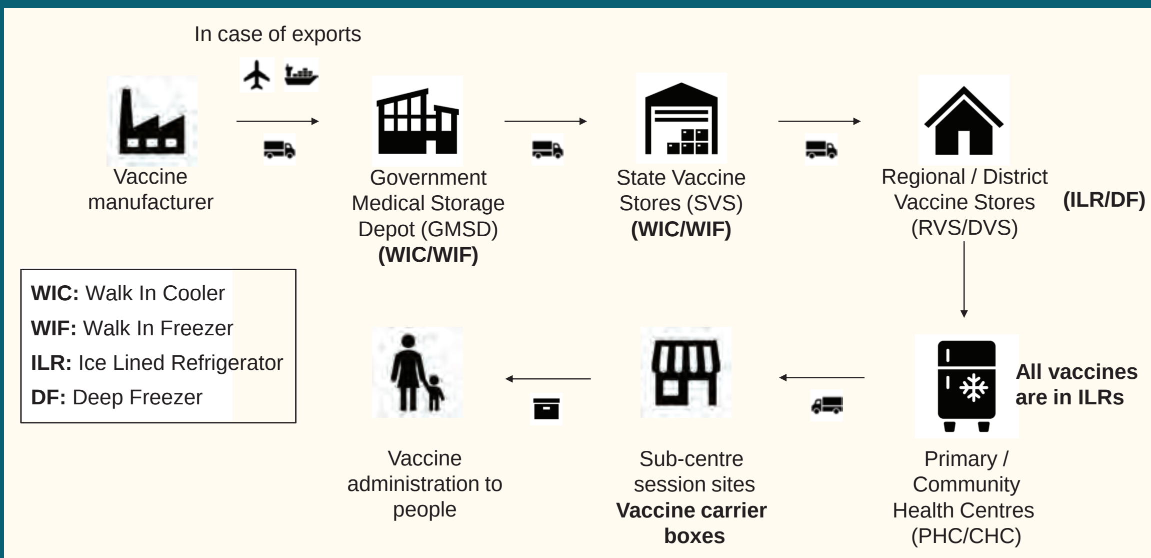
Segment	Suggested practice
Regular cleaning	Maintain a regular cleaning schedule for all surfaces. Including walls, floors, and ceilings, to prevent the buildup of contamination that could compromise fruit quality.
Inspect seals and doors	Regularly check and replace worn-out door seals to prevent air leakage, ensuring that the ripening chambers maintain the desired temperature and humidity levels.
Monitoring ethylene levels	Monitor and control ethylene levels to achieve optimal fruit ripening and avoid overexposure, which can lead to uneven ripening and quality issues.
Frequent airflow checks	Ensure that the airflow patterns within the chambers are consistent to guarantee uniform ripening across all stored fruits.
Maintenance of ripening equipment	Regular service and maintain ripening equipment, including generators, compressors, and humidification systems, to prevent malfunctions and ensure efficient performance.
Employee training	Conduct regular training sessions for staff involved in ripening operations to ensure they are aware of proper procedures, safety measures, and practices.
Pest and disease management	Implement preventive measures to control pests and diseases that could affect stored fruits. Regularly inspect and treat the ripening chamber environment as needed.
Thermal imaging	Use of thermal imaging technology to identify gaps and leakages within the facility which are usually difficult to identify with the naked eye.
Decommissioning of old systems	Regular assessment, prioritising the replacement of old equipment, modernisation, training and transition plans, proper disposal, documentation, and continuous improvement.

3

Vaccine Cold Chain

3.1. Overview of the existing landscape of vaccine cold chain in India

Figure 33: Vaccine cold chain in India [46]



The vaccine value chain adopts a top-down approach. Vaccines are transported from the manufacturer to the primary storage facilities, which include the Government Medical Store Depot (GMSD) and/or the State Vaccine Stores (SVS). These facilities represent the initial point of storage for the vaccines. Subsequently, the vaccines are distributed to the Regional Vaccine Stores (RVS) within the states, which serve to meet the requirements of their respective regions. From the RVS, vaccines are further disseminated to the District Vaccine Stores, addressing the needs of the individual districts. Following this stage, the last mile vaccine deliveries commence, reaching Community Health Centres (CHCs) and Primary Health Centres (PHCs), from where the vaccines are made available to the public.

Transportation of vaccines is conducted using either reefer transport vehicles or vehicles equipped with vaccine cold boxes and containers, which are specifically designed to maintain the temperature range necessary for the vaccines. When transported by aircraft, vaccines are stored in vaccine cold boxes and carriers, outfitted to preserve the requisite temperature range during transit.

Table 20: Vaccine temperature class and systems [47]

Temperature class	Temperature Typical range	Common storage system	Common transport system
Medium temperature range	+2°C to +8°C	Cold rooms/ compartments based on VCM technology. These can be WICs in GMSDs/ SVS and ILRs in RVS/DVS.	Purpose built unit load devices (air), refrigerated containers (sea, rail, road), qualified containers and passive cooling devices (like PCM storage containers)
Low temperature range	-50°C to -15°C	Cold rooms/ compartments based on VCM technology. These can be WIFs in GMSDs/ SVS and DFs in RVS/DVS.	Purpose built refrigerated containers (sea, rail, road), qualified containers and passive cooling devices (like PCM storage containers)
Ultra-low temperature range	-80°C to -60°C	Cold rooms/ compartments based on cascading VCM technology.	Purpose built refrigerated containers (sea, rail, road), and passive cooling device (like PCM storage containers)

Current Infrastructure

The logistics for vaccines in India has been managed through a predefined network that follows a cycle of storing and transporting the vaccine. According to the Ministry of Health and Family Welfare, there were 26,388 identified different vaccine storing points in 2016, including 53 state vaccine stores, 110 regional stores, 666 district stores and 25,555 last cold chain points such as PHC/other hospitals.

Furthermore, in 2021, India accounted for 27,000 vaccine cold chain points with 76,000 cold chain equipment and 700 reefer vans for vaccine storage.

Growth Scenario

Indian vaccine cold chain market had reached a size of USD 1.3bn in 2019[48]. There was a surge in the market size post 2020. The surge was majorly driven by the covid vaccination drive across the country and the increased demand for over-the-counter medicines. India was also one of the top exporters to low-income countries. This number will continue to rise due to the rising importance of the healthcare sector.

3.1.1. Cooling technologies

Within the vaccine cold chain, the prevailing and widely adopted cooling technology is vapour compression refrigeration. This technology has been dominant because of its ability to provide consistent and controlled cooling environments. This technology ensures that vaccines remain within the specified temperature range critical for their stability and efficacy.

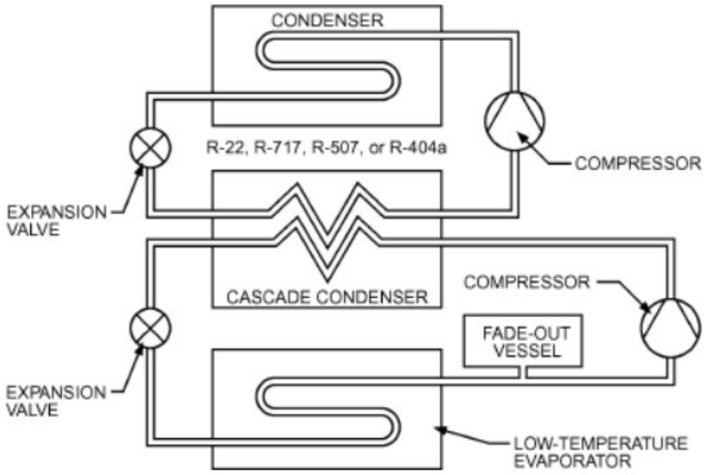
You can refer to **section 2.1.1. Cooling technologies** for more details on the vapour compression cooling technology.

For ULT (Ultra Low Temperature), vapour compression cascading technology is generally used.

ULT storage temperature is defined by a temperature range between -80°C to -60°C.[49]

A typical ULT system includes a cascade vapour compression system where the heat load from the cabinet is transferred to a high-pressure low-stage refrigerant via a cold wall evaporator design. It is then transferred to a high-stage refrigerant via an inter-stage cascade condenser, and finally to the ambient environment through a traditional air-cooled or water-cooled condenser.

Figure 34: Cascading system for ULT [50]



Another method of cooling that is adopted in the last mile delivery point is utilization dry ice.

Dry Ice: Dry ice, solid carbon dioxide (CO_2) at sub-zero temperatures, has emerged as a pivotal cooling technology in passive containers, including vaccine carriers and insulated boxes. Its distinctive characteristic of sublimation – the direct transition from a solid to a gas upon absorbing heat – renders it an efficacious phase change material (PCM)-based cooling technology. However, dry ice technology is advisable solely for temporary storage purposes.

Figure 35: Ice packs for vaccine storage [51]

Application:

- **Vaccine Carriers:** Dry ice serves as a reliable cooling medium for smaller-scale transportation of vaccines, ensuring temperature stability during transit.

- **Insulated Boxes:** Widely employed in the shipment of temperature-sensitive items, dry ice maintains the required low temperatures within insulated boxes, acting as a temporary and portable solution.

Considerations:

- **Duration:** The effectiveness of dry ice is dependent on its sublimation rate. Understanding the duration of transportation and storage helps in determining the quantity required.

- **Ventilation:** Adequate ventilation is crucial to prevent the accumulation of gaseous carbon dioxide, ensuring a safe environment during use.

Liquid Nitrogen Cooling Technology

The liquid nitrogen cooling system (LN2) involves the use of a cryogenic liquid nitrogen to achieve and maintain low temperatures. This technology can be used in two ways in a vaccine reefer system: **Closed system and open system.**

In the closed system, liquid nitrogen is electronically controlled by pressure through a heat exchanger, with an electric fan blowing the cold air through it and into the cargo area. Return fans circulate the air within the cargo area and the LN2 system to regulate the temperature desired inside the reefer vehicle.

The open system is electronically controlled by pressure through a heat exchanger with an electric fan blowing the cold air through it and into the cargo area. Return fans circulate the air inside the cargo area and the LN2 system to regulate the temperature desired inside the van.

Figure 36: Liquid nitrogen vaccine storage tanks [52]

Table 21: Storage infrastructure for vaccines[53]

Equipment	Temp. range (°C)	Level of stores	Gross storage	Storage vol. (L)	Vaccines stored
Walk in Coolers (WIC)	+2 to +8	GMSD, SVS, RVS, DVS	$16m^3$ – $32m^3$	5760 – 13440	BCG, Measles, TT, DTP, Hep B, Pentavalent, JE
Walk in Freezers (WIF)	-15 to -25	GMSD, SVS, RVS, DVS	$16m^3$ – $32m^3$	5760 – 13440	OPV, Ice packs
Ice Lined Refrigerators (ILR) - Large	+2 to +8	District & Divisional stores	198L – 218L	90 - 108	Storage of all vaccines
Ice Lined Refrigerators (ILR) - Small	+2 to +8	CHCs & PHCs	63.7L – 71L	45	Storage of all vaccines
Deep Freezers (DF) – Large	-15 to -25	District & Divisional stores	281L – 298L	200 -213	OPV, Ice packs
Deep Freezers (DF) – Small	-15 to -25	CHCs & PHCs	105L -121L	72 - 92	Ice packs
Cold Box - Large	+2 to +8	Last point delivery centres	20L – 25L		All vaccines stored or in case of power failure
Cold Box - Small	+2 to +8	Last point delivery centres	5L – 8L		All vaccines stored or in case of power failure
Vaccine Carriers	+2 to +8	Last point delivery centres	1.7L		All vaccines carried for 12 hrs (4 conditioned ice packs and 16-20 vials

Figure 37: Vaccine storage equipment [54]

> **Note:** *It is important to note that DFs should not store vaccines and ice packs together as it would lead to increase in cabinet temperature, which can be harmful for the vaccines.*

3.1.2. Refrigerant

Refrigerants play an essential role in maintaining the integrity of vaccines within the cold chain, ensuring their potency and efficacy from production to administration. The selection of suitable refrigerants is paramount in preserving the temperature-sensitive formulations of vaccines. R-134a is one of the predominant refrigerants utilised in vaccine cold chain facilities. Other commonly employed refrigerants in the vaccine cold chain include hydrofluorocarbons (HFCs), hydrochlorofluorocarbons (HCFCs), and in some instances, hydrocarbons. However, there is an increasing emphasis on transitioning towards more environmentally friendly options with lower Global Warming Potential (GWP) and Ozone Depleting Potential (ODP). Innovations are underway to adopt alternative refrigerants such as hydrofluoroolefin (HFOs) and natural refrigerants, including carbon dioxide (CO_2) and hydrocarbons. The choice of refrigerants in the vaccine cold chain is critical not only for temperature control but also for minimising environmental impact and aligning with global sustainability goals.

Please refer to **section 2.1.2.** for more details on refrigerants.

3.1.3. Insulation

Primarily used insulation is Polyurethane Foam (PUF).

Polyurethane Foam (PUF): *Overview:* For larger-scale cold storage facilities, such as those designed for vaccines, Polyurethane Foam (PUF) takes centre stage. This synthetic material, renowned for its excellent insulation properties, plays a vital role in maintaining controlled temperatures within expansive storage environments.

Application:

- **Cold Storages for Vaccines:** PUF serves as a robust insulation material for the construction of cold storage facilities, providing a barrier against external temperature fluctuations.

- **Long-Term Storage:** With its durability and efficiency, PUF is well-suited for extended storage durations, ensuring a stable environment for vaccines and perishable goods.

Considerations:

- **Thickness:** Determining the appropriate thickness of PUF insulation is essential, considering factors like ambient temperatures and the desired internal storage conditions.

- **Sealing:** Proper sealing of PUF panels is critical to prevent thermal leaks and maintain insulation efficiency over time.

3.1.4. O&M practices

In the vaccine cold chain domain, meticulous O&M practices are crucial to ensure the effectiveness and safety of vaccines. The cold chain for vaccines involves a series of temperature-controlled storage and transportation stages, and any deviation from the recommended conditions can compromise the potency of the vaccines. O&M practices in the vaccine cold chain include regular monitoring, calibration, and maintenance of refrigeration equipment, cold storage facilities, and transportation systems. It is essential to adhere to stringent temperature control measures to prevent vaccines from exposure to extremes that could affect their efficacy. The importance of robust O&M practices in the vaccine cold chain cannot be overstated, as they directly influence the capacity to deliver vaccines with optimal potency, safeguard public health, and contribute to the success of immunisation programmes worldwide.

3.2. Reefer Transport

3.2.1. Introduction

Reefer vehicles play a very crucial role in transportation of vaccines. They help in safely transporting the vaccines by maintaining specific temperature conditions. These vehicles are equipped with refrigeration units to ensure that the vaccines remain at the required temperature throughout the journey, safeguarding their potency and efficacy.

Figure 38: Vaccine truck exterior and interior [55] [56]

Practice in vaccine cold chains [57] [58]:

Both normal temperature and special ULT (Ultra Low Temperature) refrigerated transport containers are available in the market. Containers can be supplied with either passive cooling using PCMs like dry ice to keep the contents cool during transport or with active cooling if one is able to connect to an electric power supply.

This supply would use a battery package. These containers are equipped with continuous remote controls and monitoring equipment and can provide accurate and reliable temperature control through the journey. They are typically used for pharmaceuticals, biological samples, and high-grade food products.

Vaccine Van

A vaccine van is a specially designed vehicle that is used to transport vaccines in large quantities. It is crucial to ensure that vaccines are transported in cold boxes equipped with the appropriate number of ice packs. These cold boxes should be loaded into the vaccine van promptly after packing the vaccines and unloaded upon reaching the destination. Once at the destination, the vaccines should be taken out of the cold boxes and placed in the Ice Lined Refrigerator (ILR) without delay.

Cold Box

A cold box serves as an insulated container for the transportation and temporary storage of vaccines and ice packs. Its functions include:

- Collecting and transporting large quantities of vaccines
- Store vaccines up to five days
- Store vaccines when ILR breaks down as a contingency measure
- Used for storing frozen ice packs

Ice Packs

Ice packs are containers made of plastic that are filled with water and subsequently frozen solid in a deep freezer. These frozen packs serve multiple purposes, such as enhancing and preserving the holdover time of vaccines in vaccine carriers and cold boxes. They are also utilised as inner linings in ILRs to improve and maintain holdover time in the event of a power outage. In small deep freezers, approximately 20-25 ice packs (equivalent to 8-10 kg of ice) can be frozen in a single day, while larger deep freezers can accommodate around 35-40 ice packs (12-14 kg of ice). The standard ice packs used in the Universal Immunization Programme for cold boxes and vaccine carriers have a capacity of 0.4 litres.

3.2.2. Transitioning to sustainable options

Transitioning to sustainable options can be done in two cases: retrofitting existing structures or constructing whole new arrangements.

3.2.2.1. Retrofitting

Below are the potential retrofits in the following areas:

1. Insulation
2. Cooling technology
3. Other design interventions

1. Insulation

You can refer to retrofitting **section 2.3.2.1. - Insulation** for more details.

2. Cooling technologies

You can refer to retrofitting **section 2.3.2.1. - Cooling technology and section 2.4.2.1 - Cooling technology** for more details on retrofitting of cooling technology.

3. Other design interventions

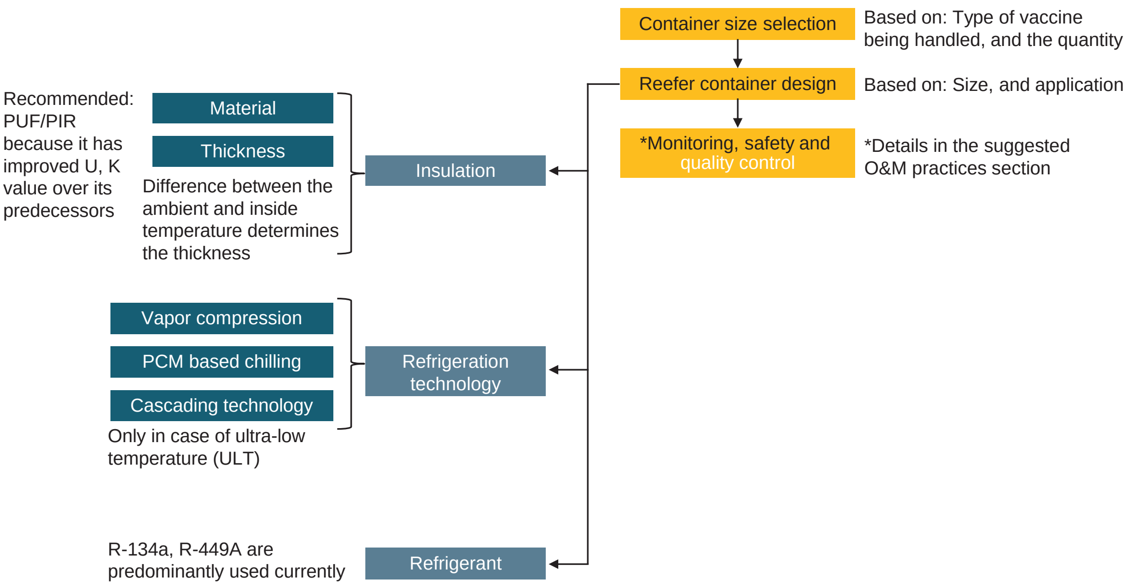
You can refer to retrofitting **section 2.3.2.1. - Other design interventions** for more details on retrofitting of various design parameters.

3.2.2.2 New Setup

Figure 39 depicts a guide to setup a reefer vehicle specialized for transporting vaccines. The process remains similar to that of perishable.

Figure 39: New setup for a vaccine reefer van

The cascading system is used in case of ultra-low temperature (ULT) vaccine



storage. Please refer to **section 3.1.1.** for more details. For more details on PCM based cooling technology, **please refer to section 2.3.2.2.**

It is advised to consult a reefer expert for designing a new reefer vehicle.

3.2.3. Suggested O&M practices

Table 22: Suggested O&M practices for vaccine reefer vehicles

Segment	Suggested Practices
Pre-loading inspection	Conduct a thorough inspection to check the overall condition of the vehicle, including tires, lights, and refrigeration unit. Verify that the reefer unit is clean and free from any debris or obstructions.
Loading and unloading procedures	Ensure proper loading and unloading procedures to maintain an even air distribution within the cargo area. Minimise the time the doors are open during loading and unloading to prevent temperature fluctuations.
Temperature monitoring and logging	Vaccines typically require more rigorous temperature monitoring with frequent logging and validation to ensure compliance with strict temperature requirements.
Cleaning and sanitation	Both perishable goods and vaccines require clean and sanitised reefer units to prevent contamination. However, vaccines may require even stricter sanitation protocols to minimise the risk of cross-contamination.
Fuel management	Regularly inspect and maintain the fuel system to ensure a consistent and adequate fuel supply for the refrigeration unit. Monitor fuel consumption and efficiency.
Insulation checks	Inspect the insulation of the cargo area for any damage or wear. Repair or replace insulation as needed to maintain thermal efficiency.
Driver training	Train drivers on proper reefer unit operation, including temperature control, defrosting procedures, and emergency protocols. Emphasise the importance of careful driving to minimise shocks to the cargo.
Emergency preparedness	Equip the vehicle with emergency equipment, such as backup power sources, temperature recording devices, and tools for basic repairs. Train personnel on emergency response procedures in case of refrigeration unit failures or temperature excursions.
Documentation and compliance	Maintain appropriate temperature and RH monitoring logs along with any other documentation required for compliance.

Segment	Suggested Practices
Roles and responsibilities of personnel involved including loader, unloader, and other crew members shall be defined	Clearly define the duties and obligations of each individual involved in the process, ensuring accountability and smooth operations.
Labelling and marking of packages and reefer trucks	Implement a comprehensive labelling and marking system for both packages and reefer trucks to facilitate easy identification, tracking, and handling of goods,
Storage, lashing and locking practices	Establish proper procedures for the storage of goods within reefer trucks, including securing them with appropriate lashing and locking mechanisms to prevent shifting or damage during transportation.
Maintenance and monitoring including preparation of maintenance schedules	Regularly inspect and upkeep the equipment and facilities, adhering to a pre-defined maintenance schedule to prevent any breakdowns.
Training for other members like loaders, unloaders, and other crew members	Provide comprehensive training programs for all the personnel involved in the process, including loaders, unloaders, and other crew members to ensure that they are equipped with the necessary skills and knowledge to perform their duties efficiently.
Decommissioning of old systems	Regular assessment, prioritising the replacement of old equipment, modernisation, training and transition plans, proper disposal, documentation, and continuous improvement.
Troubleshooting of reefer vehicles	Establish a comprehensive protocol for identifying, analysing, and rectifying potential issues encountered with reefer vehicles, encompassing challenges such as temperature variability, mechanical breakdowns, and electrical irregularities.

3.2.3 Suggested O&M practices for last mile vaccine delivery in India:

Table 23: Suggested O&M practices for last mile vaccine delivery in India

Segment	Suggested Practices
Conditioning of ice packs	Ice packs in the vaccine carriers / cold boxes should be conditioned prior to keeping the vaccines. It ensures that the vaccines remain within the recommended temperature range. Note: Conditioning of ice packs will vary from region to region depending on the temperature conditions. Stakeholders are requested to consult an expert for the same.
Regular inspection and maintenance of cold boxes	Conduct regular inspections of vaccine carriers and cold boxes to ensure they are in good condition and free from damage or wear. Any issues such as cracks, leaks, or broken seals should be promptly addressed to maintain the effectiveness of insulation.
Optimal packing of vaccine carriers	Train delivery personnel on the proper packing of vaccine carriers to maximise the utilisation of space while ensuring adequate insulation and distribution of dry ice. Proper packing techniques can help maintain temperature stability during transit.
Monitoring and replenishment of dry ice	Implement a system for monitoring the temperature and condition of dry ice within vaccine carriers during transit. Regularly replenish dry ice as needed to maintain the required temperature range for vaccine storage.
Quality control checks	Conduct regular quality control checks on vaccine carriers and cold boxes to verify their performance in maintaining temperature stability. This can involve simulated testing scenarios or spot checks during actual vaccine delivery operations.
Training on handling and storage	Ensure that delivery personnel receive thorough training on the correct procedures for handling and storing vaccine carriers. This training should encompass instructions for loading and unloading, strategies to minimise exposure to external temperature variations, and essential practices for preserving the cold chain's integrity.

3.3. Cold Storages

3.3.1. Introduction

Cold storage facilities for vaccines play an important role in maintaining the integrity and potency of vaccines throughout the supply chain. In India, GMSDs (Government Medical Store Depots), SVS (State Vaccine Stores), and RVS (Regional Vaccine Stores) are the bigger facilities storing vaccines in bulk at different stages of the vaccines journey.

Figure 40: Vaccine carriers and WIC/WIF in a vaccine storage facility[59,60]

GMSDs are central storage facilities managed by the government at the national level. They serve as the hubs for receiving and distributing vaccines to various regions within the country. There are seven GMSDs located across the country to cater to the respective regions. They are located in Mumbai, Kolkata, Chennai, Hyderabad, Guwahati, Karnal, and New Delhi.

Each state in India typically has its own vaccine store or depot (SVS) responsible for storing and distributing vaccines within the state. These facilities are equipped with cold rooms (WIC/WIF), refrigerators, freezers, and vaccine carriers.

In addition to GMSDs and SVSs, there may be RVSs (Regional Vaccine Stores) established to serve specific geographical areas or clusters of districts. They usually function as intermediate distribution hubs between SVSs and district vaccine distribution centres.

Some of the system design measures that must be followed for a vaccine cold chain are:[61]

- Thickness of PUF panel for +2ºC to +8ºC should be minimum of 80mm.
- Voltage stabiliser is a must. The power supply should pass through the voltage stabiliser and then go to the machinery in order to protect the cold chain equipment.
- The machinery, including the compressor and condenser, ought to be three-phase since the loads are more evenly balanced in three-phase systems. This results in reduced heating losses and, consequently, enhanced efficiency.
- DG set, AMF controls, regular controls with high temperature alarms and chart recorders should be in place with cold storage facility since vaccines require very stringent handling.

3.3.2. Transitioning to sustainable options

Transitioning to sustainable options in cold storages is essential for mitigating environmental impact and fostering long-term resilience in the face of climate challenge. This transition is important for maintaining a balance between meeting growing demands for refrigerated storage and minimising the environmental impact.

3.3.2.1. Retrofitting

Below are the potential retrofits in the following areas:

1. Cooling technology
2. Refrigerant
3. RE Integration
4. Other design interventions

1. Cooling technology

The retrofitting options available for cooling technology for vaccines are targeted towards GMSDs, SVS, and RVS, since these contain WICs, WIFs, ILRs and DFs. There are four possible retrofit options: Installing VFDs in a fixed speed compressor, retrofitting with a higher efficiency compressor motor, and retrofitting of atmospheric condenser with evaporative condenser. You can refer to **section 2.4.2.1 – Cooling technology** for more details on the first three retrofit options.

Retrofit option 4: Freeze preventive vaccine carriers for last mile delivery

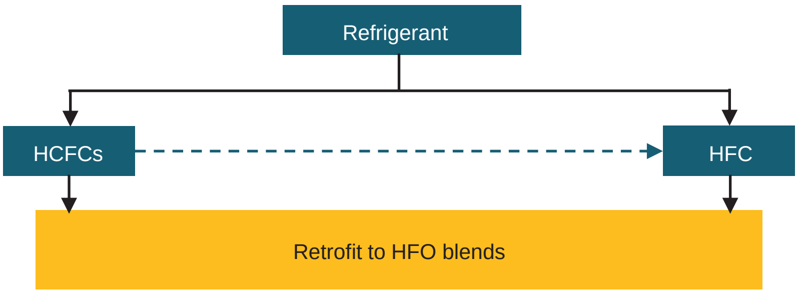
Coming to the last mile delivery cold chain for vaccines, one possible retrofit option is freezing preventive vaccine carriers.

Figure 41: Normal and freeze preventive vaccine carrier boxes [62]

Freeze-preventive vaccine carriers or cold boxes contain a barrier that separates the vaccine storage compartment from the dry ice packs to prevent direct contact. They possess a marginally lower storage capacity compared to standard vaccine carriers or cold boxes.

2. Refrigerants

Figure 42: Retrofit of refrigerants in a vaccine cold chain



In the vaccine cold chain, natural refrigerants are typically not employed. Ammonia, the most widely utilised natural refrigerant in India, is not favoured primarily due to its toxic properties and the scale of operations required, which is generally substantial with ammonia.

You can refer to retrofitting **section 2.4.2.1 - refrigerants** for more details on retrofitting of refrigerants.

3. RE integration

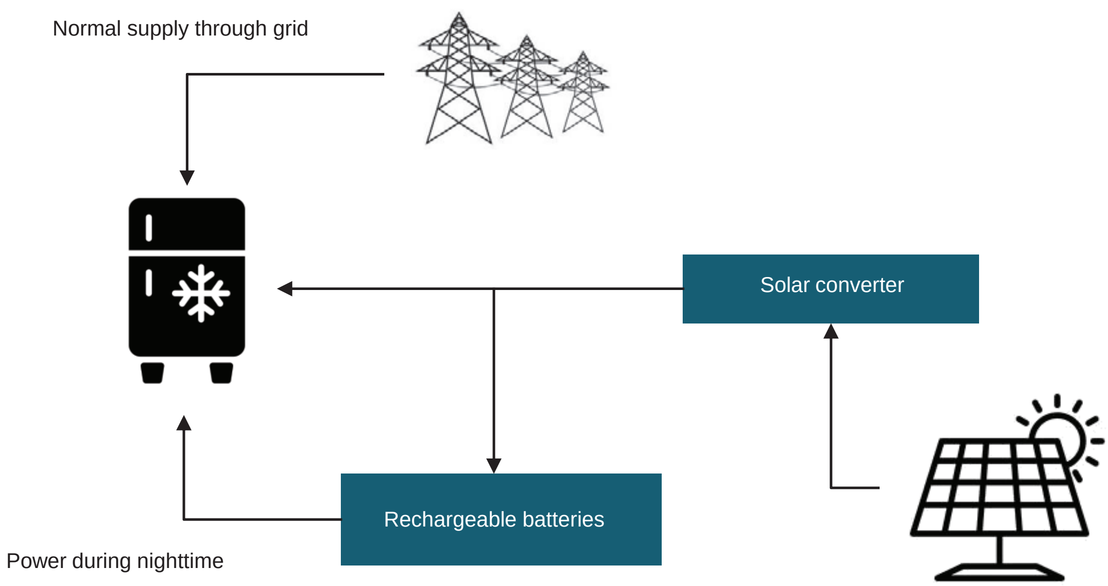
There are three potential retrofit options for solar integration when considering a vaccine cold chain. The first is the solar and grid option, typically aimed at larger vaccine stores such as GMSDs, SVS, etc., which house the WICs and WIFs. These facilities require a significant and stable electricity supply.

The second option involves solar integration with the grid, supplemented by a battery system. This approach is suitable when the grid supply is unreliable. The CAPEX will rise considerably in this scenario due to the additional cost of the battery system.

Both of these options have been discussed in **Section 2.4.2.1, RE Integration**.

The third option pertains to solar integration for smaller-capacity refrigeration systems such as the ILRs and DFs. This solution is relevant for the RVS and DVS, which contain these refrigeration systems and, unlike the WICs and WIFs, do not demand a high amount of electricity for operation.

Figure 43: RE integration for ILR/DF



Stable grid supply poses numerous challenges in rural areas, adversely affecting the vaccine cold chain, particularly in last-mile delivery. The aforementioned solar integration option endeavours to bridge the gap in rural regions or areas with unstable electricity supply.

Cost estimate (*Illustrative example*):

Table 24: Cost estimation for RE integration of ILR/DF[63]

Parameter	Details	Assumptions
Area of solar panel	24 sq. mt. (2kW)	10-12 sq. mt. / Kw
Cost per panel	INR 80,000	INR 40 per watt
Installation cost	INR 32,000	INR 16 per watt
Inverter cost	INR 18,000	INR 9 per watt
Avg. generation per day	8 kWh (8 units)	4 kWh (4 units)
CAPEX	INR 1,30,000 excluding transportation cost	
Estimated savings per year	INR 23,360	8*365*8
Return on Investment	18% (assuming best-case scenario)	
Payback	5-6 years (assuming best-case scenario)	

This serves merely as an illustrative example for straightforward comprehension. Nevertheless, it is strongly advised to consult a solar integration expert before commencing with this integration.

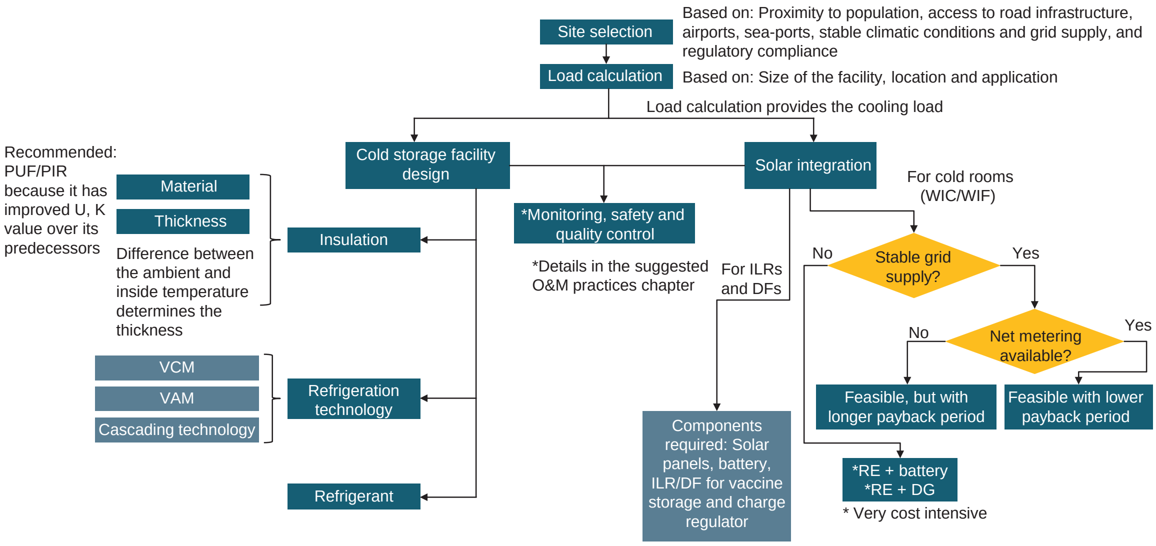
4. Other design interventions

You can refer to retrofitting **section 2.4.2.1** for more details on retrofitting of design interventions.

3.3.2.2 New Setup

Vapor compression is the most common refrigeration technology. For more details on vapor compression and vapor absorption, you may refer to **section 2.1.1.** For more details on cascading technology, you may refer to **section 3.1.1.** Solar integration is feasible for both cold rooms (WIC/WIF) and ILRs. When integrating with cold rooms, one must consider grid availability due to the substantial load these rooms consume. For ILRs, integration with a solar unit can be achieved through the utilisation of solar panels, a battery system, and a charge regulator.[64]

Figure 44: New setup for a vaccine cold storage facility



Some design aspects which need to be considered while designing a vaccine cold storage are:[65]

1. There should be adequate space to accommodate WICs/WIFs, ILRs, DFs, and racks for placing the vaccine carrier boxes.
2. A cool space not exceeding 25°C for loading and unloading of vaccines.
3. Office space with view of the entire storage zones in the facility.
4. A proper vehicle loading bay, and
5. A dry place for storage of other equipment like diluents, droppers, and syringes.

3.3.3. Suggested O&M practices

Table 25: Suggested O&M practices for vaccine cold storage

Segment	Suggested Practices
Temperature monitoring and alarm	Implement a highly sensitive and continuous temperature monitoring system specifically designed for vaccines. Set up real-time alarms to promptly respond to any deviations from the recommended storage temperatures, ensuring the vaccines remain within their specified temperature range.

Segment	Suggested Practices
Backup power system	Install reliable backup power systems, such as uninterruptible power supplies (UPS) or generators, to ensure a continuous power supply. This is critical for maintaining the required storage conditions, especially during power outrages or emergencies.
Cleanroom standards	Adhere to strict cleanroom standards to prevent contamination of vaccines. Implement regular cleaning and disinfection procedures for both storage rooms and equipment to maintain a sterile environment.
Regular training on vaccine handling	Provide ongoing training for personnel involved in vaccine storage on the specific requirements and best practices for handling and storing vaccines. Stay updated on the latest guidelines and recommendations from health authorities.
Vaccine segregation and storage zones	Implement a systematic approach to segregate vaccines based on their temperature requirements and storage conditions. Designate specific storage zones within the facility for different types of vaccines to prevent cross-contamination and ensure precise temperature control.
Risk assessment and mitigation	Conduct regular risk assessments to identify potential threats to vaccine storage integrity. Develop mitigation plans to address identified risks, ensuring a proactive approach to preventing issues that could compromise the quality of stored vaccines.
Maintenance operations	Keep the condenser unit well ventilated, conduct routine checks for appropriate functioning of WICs/ WIFs, ILRs and DFs, removal of ice accumulation on evaporators, dust accumulation on solar panels (in case solar energy is used). Clean and inspect vaccine carrier boxes after every use, ensure that the boxes are tightly sealed and packed once vaccines are stored and follow the SOPs while storage, transportation, and distribution of vaccines.

Segment	Suggested Practices
Integration of smart monitoring	A smart monitoring system based on cloud can be implemented for the components of the vaccine cold chain which will help to regulate monitoring, schedule routine maintenance, and provide alerts on any abnormalities and discrepancies.
Decommissioning of old systems	Regular assessment, prioritising the replacement of old equipment, modernisation, training and transition plans, proper disposal, documentation, and continuous improvement.

Endnotes

1 ICAP - 2019

2 Bureau of Indian Standards

3 https://www.araner.com/blog/vapor-compression-refrigeration-cycle

4 http://home.iitk.ac.in/~samkhan/ME340A/Lecture_14_Vapor_Absorption_Refrigeration_OK.pdf

5 Stakeholder consultations

6 https://www.hrai.ca/uploads/userfiles/files/refrigerant_table_June2019.pdf

7 https://theinsulationsuppliers.co.uk/our-products/insulation/polystyrene-insulation-premium-eps/#:~:text=Typical%20U%2Dvalue%20for%20standard,board%20is%200.038W%2FmK%20.

8 https://www.rmax.com/blog/eps-insulation-r-value#:~:text=A%20typical%20EPS%20insulation%20R,will%20have%20a%20minimum%20R8.

9 https://classiccomfortohio.com/pdfs/Foam-Control-EPS-Water-Absorption-Facts.pdf

10 https://polyfoamxps.co.uk/what-are-the-similarities-and-differences-between-extruded-polystyrene-xps-and-pir-insulation/#:~:text=An%20XPS%20insulation%20product%20usually,of%20around%200.033%20W%2FmK.

11 https://www.ixl.co.th/en/products/ixl-insulated-panel/xps-extruded-polystyrene-or-styrofoam/#:~:text=XPS%20(Extruded%20Polystyrene%20or%20Styrofoam)%20has%20a%20standard%20density%20of,50%20to%2075%20degrees%20Celsius.

12 https://www.sf.technology/en/productDetail?id=22719d256e30bf3d016e342c04e60009

13 https://www.poliuretano.it/EN/thermal_conductivity_polyurethane.html

14 https://highperformanceinsulation.eu/wp-content/uploads/2016/08/Thermal_insulation_materials_made_of_rigid_polyurethane_foam.pdf

15 https://www.energy.gov/energysaver/vapor-barriers-or-vapor-retarders

16 https://www.indiamart.com/proddetail/cold-room-flush-doors-7719748012.html

17 Stakeholder consultations

18 Stakeholder consultations

19 https://www.mjtrucknation.com/future-of-reefer-trucks/

20 Stakeholder consultations

21 Stakeholder consultations

22 Stakeholder consultations

23 https://www.dsu-nl.com/en-EU/fridge-seal-is-broken-or-worn-out

24 https://www.coolingindia.in/cold-chain-logistics-and-refrigeration-using-pcm/

25 Study on cold chain sector in India for promoting non-ODS and low-GWP refrigerants – Ozone cell, MoEFCC

26 Study on cold chain sector in India for promoting non-ODS and low-GWP refrigerants – Ozone cell, MoEFCC

27 https://www.coolingindia.in/design-of-cold-storage-for-fruits-vegetables/

28 Mecalfab Ltd.

29 Stakeholder consultations

30 Stakeholder consultations

31 Stakeholder consultations

32 https://ekeindia.com/applications/motor-efficiency-iec-standards-classification/

33 https://ekeindia.com/applications/motor-efficiency-iec-standards-classification/

34 Stakeholder consultations

35 Stakeholder consultations

36 Stakeholder consultations

37 Stakeholder consultations

38 https://www.ceew.in/publications/how-indias-solar-and-wind-policies-enabled-its-energy-transition

39 https://www.ceew.in/publications/how-indias-solar-and-wind-policies-enabled-its-energy-transition

40 https://www.facilitiesnet.com/energyefficiency/tip/Loading-Docks-Seals-Can-Improve-Energy-Efficiency--37042

41 Stakeholder consultations

42 Gandhi Automation Pvt. Ltd.

43 Gandhi Automation Pvt. Ltd.

44 Stakeholder consultations

45 Technical standards for and protocol for the fruit ripening chamber in India - NHB

46 Handbook for vaccine and cold chain handlers – 2nd edition, 2016

47 Practical guidance for vaccine refrigerated transportation and storage – ASHRAE

48 https://www.iata.org/en/publications/newsletters/iata-knowledge-hub/vaccines-and-the-india-market/

49 Practical guidance for vaccine refrigerated transportation and storage – ASHRAE

50 Practical guidance for vaccine refrigerated transportation and storage – ASHRAE

51 Handbook for vaccine and cold chain handlers – 2nd edition, 2016

52 https://www.mediproducts.net/blog/vaccine-storage/liquid-nitrogen-freezer-or-dry-ice-for-vaccine-storage

53 Cold Chain & Logistics management by WHO & PQS (WHO), In-depth analysis of cold chain, vaccine supply & logistics management by INCLEN Trust International, supported by Ministry of Health & Family Welfare, Govt of India.

54 Handbook for vaccine and cold chain handlers – 1st edition, 2010

55 https://topolo-truck.com/medicine-reefer-truck-body/

56 https://weather.com/en-IN/india/coronavirus/news/2021-01-16-heres-how-india-plans-to-store-covishield-and-covaxin-vaccines

57 https://ozone.unep.org/system/files/documents/TEAP-RTOC-technical-note-vaccines-cold-chain.pdf

58 https://main.mohfw.gov.in/sites/default/files/Unit4ColdChainandLogisticsManagement.pdf

59 https://www.undp.org/india/building-home-vaccines-chhattisgarh

60 http://www.nccmis.org/document/Final-WICWIF-Repair-Maintenance-Training-Module.pdf

61 Stakeholder consultations

62 https://www.who.int/publications/i/item/WHO-IVB-2021.02Rev.1

63 Stakeholder consultations

64 Handbook for vaccine and cold chain handlers 2nd edition 2016 – Ministry of Health and Family Welfare

65 Training module – Walk-In-Cooler/Walk-In-Freezer Repair and Maintenance

SEPTEMBER 2024

WHOM TO CONTACT TO LEARN MORE ABOUT OZONE

Ozone Cell
Ministry of Environment, Forest and Climate Change
1st Floor, 9 Institutional Area, Lodhi Road, New Delhi - 110003
P: 011-24642176 | F: 011-24642175 | ozonecell.nic.in